

2nd Applied Active Inference Symposium on "Robotics" ~ 2nd session

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Duration: 4:00:19 | **Uploaded:** Unknown **Transcript method:** auto_caption

[Music] all right hello and welcome everyone welcome back to the second applied active inference symposium it is the second interval and it's july 31st 2022 in this second session we are going to first feature a presentation by bruno lara prediction error dynamics a proof of concept implementation following that we'll hear from matt brown with a presentation real-time robotic control through embodied homeostatic feedback the third presentation will be by adam saffron on generalized simultaneous localization and mapping gslam as a unification framework for natural and artificial intelligences and the final presentation will be by jf klochter towards a symbolic implementation of active inference for lego robots we'll then have some of those presenters rejoin us in the final two or so hours of this interval for a round table discussion we'll also be joined by carl first and there so thanks everyone for watching live or in replay hope that you add any comments if you would like into the live chat and otherwise thanks to all of our presenters and co-organizers for their awesome work and thanks so much for joining bruno please take it away yeah thanks a lot for the invitation um i was very surprised uh as as most of our work so far it's not directly related um well it's not actually active inference of course it's related um but um since we published a couple of years ago a paper well a paper that actually i'm pre i'm presenting a bit of this results here well lots of lots of nice and interesting people started to look into this this implementation of of prediction aerodynamics and it's actually the the part where where um we could say it's a part that we are taking out of all this framework um to try to implement in our in our robots in our agents for for i don't i i i don't know how much or how many times you've heard this story of of why we are doing this kind of cognitive robotics or why we like to call all this all this research or this area cognitive robotics and i like a lot this slide which kind of summarizes what it's usually a long introduction on the history of artificial intelligence and robotics and it just tells this this story of 50 years of research trying to figure out how can we have machines that are as intelligent as humans and it comes out that actually this this upper part of the image quite good

resolve rather fast we cannot say easily but but it was more or less fast and then we have the second part of the slide like the bottom part which is what comes out to be the most difficult challenge for robotics no so what has to do with interaction with the world you have a small child that is capable of handling this these objects in the environment and then you have this other robot that can do it but at the time the moment that you change the size the weight the position whatever you want of the of the pieces or of the chessboard and you actually have a child that with no problem can adapt himself to to the challenge you know and and and the the nerd joke is then on the other side you have another phd that has to solve it if you do the same for the robot you know so in reality what was supposed to or was thought to be kind of the the heart the heart problem which was having a robot reason and and and solve problems it came out to be quite the the not so difficult to solve tasks and and we can talk about uh more difficult challenges that have already been passed such as i don't know playing go which is way harder than than chess and we had alphago a couple of years ago um well winning against the human the human champion so we like to think that we are we are doing now uh things differently we are having robots uh that interact with the world that interact with what's going on around and and they learn through this interaction and of course that gives us lots of challenges and lots of things that we have to solve now i come from from from this yeah you could think now as an old school of internal models um uh we we we work in in this um in this field for for quite quite already some years and and it's this this this part of the world where there is inverse and forward models for control of course and and they are they are thought to be behind all these all these important issues in in humans and since since the beginning of my career we've been doing implementations of different types of of of tools for for these robots but always based on these inverse and forward models so just just a quick reminder for for those that might not be very very much into this this type of models so we had a sensory situation whichever it is whatever it is and we have a desired sensory situation we have an inverse model that takes as input does those two things the the the sensory situation at time t of course the sensory situation at time t plus one which is the desired and the inverse model is going to tell us uh what's the motor command that needs to be performed to go from here to here right and then we have a forward model that takes the sensory situation at time t and the motor command at time t and gives us the prediction of what's going to happen with this sensory situation and so we have a prediction error when we well we have lots of errors but um we we are interested on on particularly this um error which is the difference between what we actually predicted and what happened in the world we've been doing this for

for quite a while uh um well we've also used this this this prediction back as a sensory situation and we can get long-time predictions what we called we can do lots of comparisons between what we wanted and what actually happened and lots of different errors but as as you know and well now everyone works with is actually this error like the difference between the world what actually i predicted yeah and so that's that's that's what we can say we have in common some years uh after after well some years ago actually we came across all these all these ideas of prediction uh producer reproductive processing and we started to to to dig into it to see what actually we could use for our models what what could be useful for for improving our models and our our uh our agent's behaviors and uh we have lots of doubts lots of lots of questions but one of the first things that we've done so far is to look into what happens with the error as as an agent is executing an action sorry but first i i'd like you to walk you through one of our our implementations well first where where is actually the um what you could call the the the crowds uh areas of what we believe it's it's important in this case when we have active inference and prediction error minimization most of us are very familiar with this kind of um representation of of what's happening during during action uh and we actually believe that prediction a minimizing prediction error is what leads to belief updating so like the change in a prior belief encoded as a posterior belief and this is actually informational gain so this means learning right in active inference the expected futures their states are fulfilled through action execution we know and this associated expected prediction error is minimized by actively sampling sensory information and so the the important part for us is this expected prediction error and we believe the attribute of a policy which can be epistemic and instrumental in in case this is this action is novelty searching it's reflecting its epistemic affordance and we could say it's acting for information uh in case of preference searching no this reflects instrumental affordances and its action for reward so in words of of kirbenstein we can we can think of active inference as the process of selecting relevant affordances so those which are expected to minimize prediction error in a context-sensitive manner yeah what we've done actually or what we wanted to to actually implement uh is how these how these um expectations about the the the dynamics of the error um cause or are related to feelings to to to these embodied feelings and we believe they they allow a sensitivity so how well or how bad an agent is doing at improving the grip on what is relevant in the landscape of affordances so the sensitivity to changes in the rate of prediction error reduction is what we call a prediction aerodynamics we can say the steeper the slope the faster the rate of reduction and we see some some like uh yeah examples just just very uh very simple so in in red in in black we could see

the actual error as as like as the action is executing and then we have a slope associated to this error so what we don't want what the agent doesn't want is this error increasing having a positive slope on the other hand the error is decreasing then we have this slope and this is actually what we what we are looking for so prediction errors are manifested as changes on affect so if the rate of prediction error is faster than expected we have a positive balance and we have an action policy which is more precise uh one in the case that the error is slower than expected or is increasing unexpectedly we have a negative violence and an action policy which is less precise just just walking you through a couple of implementations that we have done the first one is just kind of an example of of this mix that we are trying to have the error is used for learning but we have a very simple um implementation of self-organizing maps we call it the self-organized internal models architecture the the nice thing on this architecture is that it actually sort of doesn't have to have an inverse model what we have is a an um self-organizing map that organizes the motor sensor information and on the other hand we have a visual sensory information which is also organized in the same another self-organizing map and then we have what we call a multi-modal representation and in this multi-modal representation we have associations we have associations between a sensory situation another sensory situation and a motor command so you could think of this as an inverse forward model pair but it's actually not encoded as such so we just have associations between sensory situations and motor commands and so we we thought this was very interesting and we use it for example for psychotic control so the robot has some initial sensory situation which is the stimulus somewhere in the in in its in its uh field of view and what we need or what what the robot wants is to phobiate the stimulus no so this is the sensory situation the desired one we ask this self-organizing maps architecture what we need to do so to bring the stimulus from here to here and it's executed and it's not very precise so it comes close to the center of the image and then we use this as the new sensory then we have the same target we again ask the architecture what what do we need to do and then it comes very close so after 100 runs you can see that the robot this is the starting situation in red we have the prediction and execution and in green we have the second prediction and execution so it's it's it's a very nice simple architecture that actually works as i said before it works like a forward model but it can also work as an inverse uh we use prediction error to learn we we could use it as some kind of modular um we can put more maps and and in this case we have the position of the head visual stimulus the position of the arm and what the robot can actually do is find the the the stimulus somewhere in the space and then bring it to the to the center kind of obviate it and

then actually bring the arm also there so we have all these associations between movements of the head movements of the arm movements of the eyes and all these get associated and are coded as initial and ending positions okay so the next question or the next thing we tried to do was how can we use prediction aerodynamics to guide learning and and this was this paper we presented a couple of years ago in in epirope was actually guidos kilachi that presented it which is our co-author so it's a work um well by guido alejandra syria which is also a colleague and myself and what we did there is um i i won't uh work on the on the i won't tell you much about the the maps or the or the kind of networks that were used and in detail the important part is uh what we see here we have a very simple robot this is what you could see here it's a 2 two degrees of freedom um camera mounted on on a sort of artificial field uh where we have lots of lots of plants the the the robot moves towards a specific destination so it looks it wants to look at some at some image at some point and so we have for each movement for each goal sorry for each goal we have a prediction error so we organize the goals in a sum of course the goals sorry are encoded using a an auto encoder we use the middle part of the encoder to have represented the goals here and the important part is each each one of these goals has some prediction error dynamics we go back again to our inverse model and forward model and execute the command the motor command and we get our we do our prediction and we get an between what we wanted and what happened the interesting part and what became kind of nice and that was what we wanted to implement what we were thinking about is we append this prediction error to as a sort of a buffer this buffer once that it contains more than four predictions we do a linear regression and actually what we get is an indicator of the progress uh how good are we doing so it's kind of a tendency for for for for learning so when we have a negative slope it actually means we are going towards the goal we are doing fine progress is increasing and it means there is a positive emotion and we remain on this goal so learning is going fine when the when the slope is positive so the activities away from the goal there is no progress the error keeps increasing so we have what we could see as a negative emotion something that can be encourages them to switch to switch goal so we have a positive slope means increase the size of the prediction error dynamics buffer this was one of sorry that was uh something of kind of interesting and and very uh yeah new in in in in this in this field in this area we haven't thought about it before and what we actually did is okay what happens if if my if my slope is positive it means i'm doing very bad at this task and what i'm going to do is i'm going to increase the size of this buffer when when we have a negative slope it means okay i'm doing fine so i'm going to decrease the size of so what we are actually doing is

adjusting dynamically the size of this monitoring window and and we're doing that depending on how things are going with respect to what we are actually expecting so if the prediction error increases we have a positive slope and we do a more frequent evaluation of the feedback this allows a quick correction of the action it can lead to satisfaction in case where we managed to to do it better and induces abandoning the goal so after in this case after a number of iterations interactions uh where the error is not decreasing the error keeps increasing then we can change we can switch goals and we could say we are trying to avoid frustration through intrinsic motivation what we try is search for a new goal when the prediction error decreases we have a negative slope we don't need to to be evaluating the feedback so so frequently and we can free resources for the agent to do something else there is no more prediction error so that means we've achieved our goal so learning happen learning occur again we search for a new goal now when we select a new goal we give priority to those that have a negative slope which is kind of a steep in the prediction aerodynamics and we have some examples here of how the error moves as we are moving in in in in the world and how the buffer size increases when the error increases and then it decreases uh how when when the slope now again changes so that was that was one of the first things we did and now actually adam invited us to this to this uh special issue that he did in frontiers and um we we published there a paper which is actually well at the beginning it was about something else and then we ended up with this with this with this model presenting this model uh this is this is just uh an idea that which actually encloses this part that i talked about about the slopes and the expected error so what what we are what we are suggesting here is that i don't know how much time we have if i can go into detail on this but um let's let's try something very fast so we have an agent that has maybe physiological needs in case the agent doesn't have physiological needs through intrinsic motivation we want to do something new we want to do search around the world then we go to what we call the environmental context which brings us to this field of affordances uh where there is loads of actions and each action each each task has some expected expected aerodynamics we go to a task related context which comes from selecting the task and then we do some planning of sensory motor sequences we start acting on the world and we get some error as we execute this task then we have two types of monitoring we have the typical um momentarily say error that helps us go through the planning and correct the the execution of the task and we have a a window of prediction error dynamics which is compared to the expected error that this section had and according to that is that we are going to go to these emotions that tells us no the the error is increasing we have this slope here so we go back to

another task uh before like of course checking then if something has changed we still have physiological needs or we don't if the error is not so greatly increasing it's more or less going fine then we go back to this same uh planning to the same so it's it's a very uh very very nice uh model that we thought of it's it's it's it's on the it's on the make and we have many ideas uh from bringing this box diagram into into the world and and in our in our idealistic uh dream what we are thinking on is something actually based on soymag we haven't implemented this yet but it's something based on this oima but now we have lots of other modules we have uh some interceptive internal model representation extra receptive um model representation and this is actually wrong but anyway so in this site we have all the receptive information the interceptive information and we have some policies and a task and some error this error is going to tell us how we are doing how we can stay on the task uh if the error is too big then we can change tasks so it's a bit of a mixture between the the the previous thing i told you and and this is how it could look in what happens on a top down so when we have a task this task is going to activate some policies and these policies are going down to act on the world we have again again uh some error coming up and this error is somewhat somehow coded here there is some monitoring so that we can go on in our policies this is the momentanus error if the error is very big then we can go back here ah sorry so this is top down and and this is actually bottom up no it's actually the error coming up all the way and coding here and uh yeah and this is how we start the cycle with some task policies and everything else that is happening in the world so i think that we can we can we can stop there i hope it's it's clear we don't have direct questions from the audience as i'm not seeing anyone or i'll ask you uh one question otherwise no we have no direct questions but we can um definitely accumulate them for the round table what is the relationship between prediction error minimization and free energy well i i think it's just like part of the process uh uh i i don't know what you mean i mean within the whole thing of of whatever you want free energy um everything that is happening the difference between the expected uh states and what actually happened in the world so we always have in in inactive inference or in free energy you always have an error happening in in what's happening what's in what's yeah what is happening in the in the action of the in the world through this prediction error is how you are we we believe it's going to happen like the learning is going to occur i don't know i don't know if yes that's awesome well yeah yeah thank you very much hope to see you very soon yeah we'll see you like in in two hours or so excellent see you soon yeah okay thank you very much hope to all right welcome everyone we are back this next presentation is going to be by matt brown real robotic control through

embodied homeostatic feedback so matt thank you very much for co-organizing and for also joining to present and uh yeah thank you uh for having me here and that was a really great talk renauld um i have some follow-up questions for later um but yeah thanks for coming and uh my name is matt i'll be taking talking a bit about um a long-running research project and uh i hope you guys find this interesting um this here's a quick high-level description of the project project um it's kind of an alternate way of looking at the generative modeling process built around homeostasis from the bottom up uh this approach is very different a bit of an orthodox approach and i'll be going into a bit of detail about how this works um you could probably read it but the short version is based on theories of ross ashby and his homeostat and it's a new kind of adaptive model that uses uh learns via homeostatic feedback through an environment and it's uh and itself uh and i'll be showing some demos about applying this to robotic control uh that i've been working on lately uh just a quick agenda um i'll be talking a bit about uh the project background uh where it came from because it is kind of unusual i figured some background might be useful uh then i'll be talking about um how homeostats uh build network self models um and what that means and how those function and then i'll be talking about plugging those into environments uh and how embodiment works and how synchronization functions uh in this process uh and then i'll be doing some uh demonstrations uh showing this working with robotics um yeah so just to give you background on myself uh i'm a computer scientist mostly i'm not much of an academic i have a few published papers but not on this particular topic um i've been obsessed with the mechanisms of thought for most of my life um i i've worked in the video game industry for about a dozen years uh and then worked in deep reinforcement learning for a number of years um and also uh about 20 years ago was uh worked in software radio uh at the spectrumware project which became genie radio that'll be relevant later um today i'll be talking about uh work that's uh now going on going at my uh startup thoughtforge and talking a bit about the history uh will help explain what i'm doing i think uh this started about uh 2006-2007 uh with um deep learning was kind of new and was kind of uh the new hot thing i did a big deep dive into it really excited about it uh but i was in fact quite uh dissatisfied and i felt that um deep learning and the this approach of using neural networks didn't really provide much insight to how learning works in in natural systems uh and even in the last few years i feel like we have this amazing hammer with deep learning and everything suddenly uh looks like nails uh and i feel like we need to take a step back and look critically at what we're what problem we're trying to solve so for me uh i was looking for new approaches in the nature of uh agency and goals uh goals in terms of what what are they and how do they

manifest in nature um semantic information versus raw information how do brain sensory signals become meaningful uh and also uh i wanted more insight into how time played into this learning process uh and i was in particular looking for media agnostic approaches um not neurons specifically how do plants in single-celled organisms um deal with their world uh so i started by reading a lot um uh going back in time chaos theory i was a big fan of i ran into matarana and varela and autopolis and i was really excited about that but it was too abstract for me as an engineer i think to tackle and so i kept going until i got to homeostasis and cybernetics and particularly a cyberneticist named ross ashby um and i basically became obsessed with ross ashby and you know to still am to a certain degree uh and he's not a part of most computer science programs so um he's quite well known outside of computer science but i felt um is not part of my own uh education uh and he's most known for things like the law of requisite variety and the good regulator theorem uh and those will be common themes um uh in the next 30 minutes or so um so what i started doing is pouring through ashby's technical journals recreating his work in a particular device called the homeostat um so this is just a quick slide to tell people about uh the homeostat in aspic uh no doubt many people watching this probably are familiar with ashby and perhaps the homeostat uh but it's worth going over briefly regardless um if you look up on wikipedia it'll say homeostat is one of the first devices capable of adapting itself to the environment and it exhibited behavior such as habituation reinforcement and learning through its ability to maintain homeostasis in a changing environment um ashby viewed uh you know learning as adaptation and stability um and was described as an isomorphism generating machine and most importantly for me um it uses feedback to generate homeostasis via a property called ultra stability um so i'll be talking about that here there's a description here from one of his books about it uh but um it is quite tricky to reproduce i had to go to his uh personal technical journals um and lots of iteration before i got it working um the key for me uh was ultra stability which is a very strange kind of property but it is the individual node property within this four node homeostat that enabled this general property this group property of homeostasis to take place uh and it's part of the function of how oops let me go back uh it's part of the uh function of how the homeostat works uh and it's in the property implemented by individual nodes uh using a simple set of rules that gives rise to group homeostasis uh it's comprised of a double feedback loop where like the low level feedback loop performs like a linear transformation and then a high level feedback loop modifies this linear transformation uh and is designed to seek equilibrium meaning output of zero which is you can if you think of the um this each of these nodes as

seeking um uh uh uh equilibrium of a state of zero um every choice of a linear transformation is a prediction of a type and so uh the result of this uh linear transformation every at every moment uh accumulates prediction error uh as uh as it seeks a zero prediction error so um it's kind of an interesting goal driven process that uh works through multi uh group process um this is just a more in-depth description of uh the homeostat basically constructed of four individual ultra-stable nodes um and the network as a whole creates an adaptive dynamical tractor set as a self-consistent reinforcing self-model ultimate stability powers its race of states disabled states space space exploration uh through like the law of representative variety uh and can be seen as a network of uh reciprocal reciprocal constraints this is just an interesting letter uh from uh uh alan turing to ashby uh back when they were members of the ratio club um and ashley was getting some notoriety um hearing um basically disagreed with ashley's approach and you know this letter here he suggests that ashby simply uh so uh simulate his homeostat in his uh tiering machine um and while true uh i think this uh actually was a bit taken aback by this um because it kind of misses the point um at the time the debate was all about analog versus digital computation um and you know uh what was useful to the war effort uh was most critical um but there was a bit of a conflation here i think in terms of what aspect was trying to create which was something more along the lines of universal adaptation or universal cognition maybe um versus turing's universal computation uh and you know uh i feel like this conflation of issues still haunts us today with deep learning and whether or not they're intelligent um so how do you build them so uh essentially i took tyrion's advice and simulated um and particularly uh 20 years ago as part of the spectrum wear project which uh you know kind of pioneered software radio uh which is now part called geniu radio uh and so i kind of stole methodologies from this approach where i you know i created the homeostat as a digital simulation of a continuous analog system or device um and so that's kind of the approach i took um i started by pointing through ask these technical journals um you know and experimented with different implementations and approaches of his direct analog machine um once this was functional uh it was simple enough to abstract down to very simple kind of minimal process um once it's functional you can start taking pieces out and see what what breaks it and what doesn't uh and eventually this uh effectively simplified down to a dynamical attractor definition for ultra stability uh that could be connected into arbitrary network topologies uh and the network as a whole can be kind of seen as a self model uh each node is trying to protect its neighbor as it seeks equilibrium um and the local ultra stability tunes nodes local connections and the resulting positive and negative feedbacks loops that it is

personally a part of and ultra speedy tunes these as a group of these overlapping feedback loops such that they reach the homeostatic equilibrium uh and that means that the uh local prediction accumulated prediction error drops to zero as well the as well as the output uh uh from each node uh and these overlapping loops become self-reinforcing and self-repairing uh and it's uh this final self-predicting set of loops that you know uh maintains a self-model uh at homeostatic equilibrium this model converges and considered self-consistent um it is composed of um these homeostatic feedback loops that maintain the network at a critical state uh meaning that um the network is highly sensitive to prediction error i mean the smalling the smallest prediction error can trigger global structural changes amongst these feedback loops and this process can also be seen as synchronization which i can talk about uh a bit later um so let me show a quick demo i'm gonna skip the original homeostat here and i'm going to um uh show a quick demonstration of what this actually looks like for a 20 node homeostat it takes a bit longer and it's a little more demonstrative so let me let me switch briefly here and apologize for the delay while i get this up cool all right so let me just describe what you're seeing here uh you're seeing 20 single dimensional graphs you could uh this is the accumulated local prediction error at each node of the 20 node homeostat you can also think of this as one 20-dimensional uh dynamical attractor or 20 single-dimensional uh dynamical tractors these are also kind of you consider these as phase space graphs um for each of the nodes uh so what you're seeing here is i have a pause i'm going to move it along slowly this is slowed down uh on purpose uh and what you'll see is they start off very chaotically um and you know what'll happen is a couple nodes will start stabilizing uh then more will start stabilizing and then eventually uh the entire network will converge to zero prediction error now this is unconnected to an environment so it's just predicting itself uh one thing i like to do here and i don't know how many people else are going to find this interesting i want to show a little in-depth look at what this final convergence and collapse to zero looks like um so um what i'm going to do is i'm going to change this to auto scaling the the graphs meaning that as it goes to zero you'll see the the the graph ranges drop uh and uh and what you'll should see is this uh usually there's a pattern forms amongst the output nodes um that is kind of consistent with every drop what you'll see is that you know um it's kind of hard to see with 20 nodes especially i've gone a little too far we're already at e negative 20 here um but what you're seeing is you're seeing the 20 dimensional drop to zero and there's kind of like a cyclical pattern to this uh and every time it goes through this pattern it drops down another order of magnitude and this will go all the way down to uh the floating point error of my computer

which is about negative 46 i believe uh and then it basically fails to function at that point because it can't tell the difference uh floating point error uh it can no longer function and the whole thing kind of falls apart there um but um that kind of shows you a bit about the process um let me go quickly back to the uh the slides so yeah that just kind of showed you a slowed down version of the 20 dimensional homeostat and i want to show the 20 dimensional because it happens a lot slower and it's a little uh easier to see what's going on with four nodes it's almost instantaneous and it's hard to see the progression um okay um so that seems like a lot of work for a model that doesn't really do anything is a connective environment so you know we have this model that can adapt this internal structure to reach homeostatic equilibrium through kind of a self-synchronization process um how do we use this to drive behavior and you know this next step took me a few years uh even though it's kind of conceptually simple um but don't worry it'll uh it'll get complex really quickly the short version is sensors and motors kind of define how the agent is embedded into an environment the network topology defines the adaptive capacity of the homeostatic agent's model uh sensitive definitions provide a mechanism to influence the behavior of the system by providing uh priors in the form of preferred environmental states these act is kind of like tela dynamic priors uh externally defined dynamic set points for the homeostatic adaptation um the motor definitions provide clinical affordances for modifying the environment that provides the basis for future sensory predictions uh the preferred states of the sensors become the preferred outcomes of the motors and the model's behavior is biased as as desired this is probably most people on this active inference symposium are probably familiar with this idea uh this slide is put together um for the benefit of people who are kind of familiar with traditional deep learning and neural network approaches to contrast how these networks function differently because it's a bit hard for people to get their head around um the first it is probably the most obvious is there's no back propagation here uh only ultra stability which is kind of a local rule um this uh leads to huge sample efficiency and uh the learning is very structured meaning you don't have to you know diffuse learning across you know thousands or millions of parameters you know you only have to model the part of the uh you don't have to update the part of the model that is mispredicting or you know is part of a misprediction um the other thing to notice is that um this is not a temporal most uh neural networks and you know like deep learning networks are kind of disembodied atemporal processes um this is very much a temporal process uh you know signals take time to move from one node to another which makes it inherently the whole thing very temporal uh the resulting temporal relationships are important actually um spatial

relationships of sensors and motors end up getting encoded into temporal relative temporal relationships within the sensory signals um such and also vice versa signal timings of the propagating signals within the network can translate into different physical spatial relationships amongst the output motors uh for example or the input sensors um so there's a relationship um between uh temporal location relative timing of signals and the spatial location of sensors and motors and the physical embodied agent um different in the you know just to kind of you know push on that a little farther the different regions within this larger homeostatic network will then be sensitive to different sets of motors and sense of sensors um based on their physical uh relative location um and uh by contrast you know you know deep learning deep enforcement learning uh deep reinforcement learning for example is an attempt at adding temporality to an otherwise a temporal static uh deep learning models um you know the other way to compare these two is you know deep learning models are generally trying to be general function universal function approximators uh in uh where this is much more uh generative and inherently temporal uh the very learning is very very different um and uh just to expand a bit more on the topics from the previous slide um the network diagram here on the right is very oversimplified uh network and it's just a meant to visually explain how to think about some of these networks uh there's no forward or backward no front back top bottom um only kind of relative relative location from other nodes and from sensors and motors perhaps more relevantly um and it is um you know common for people to look at the learning process as a sequential flow of processing like sensing perceiving then acting uh or plan planning than acting uh and instead you know you have to kind of think of this as all happening at the same time merging together is kind of one whole perceptual motor system uh not a traditional uh sandwich uh model time simply advances and we simulate the analog systems as signals propagate across the network uh and so you know you don't get input from the sensors calculate the network and get alpha motors you get you send them input to the sensors you get out from the motors base and those motor actions are basically based on the generative model's prediction of what it should be doing already before having it even sensed the world um the you know what's interesting here from the model point of view in terms of how this homeostasis works is the network's homeostatic feedback loops come to be dependent on the expected sensory signals uh meaning that the the the sensory signals become entrained in the homeostatic self-maintenance processes um which kind of naturally makes it grounded in the body experience this uh and this you know homeostatic self-maintenance process is composed of the environmental signals themselves as part of its predictions and expectations um

expectations and predictions for the consequences of motor actions are embedded in the signal sent to the sensors from deeper in the network like on the right here and uh they're sent to the sensors where expectation meets reality and prediction errors cascade backwards uh and modify the internal structure of the model uh as polynomials model update um these and then motor actions are accompanied by these parallel sensory prediction signals that flow uh from the deeper in the network uh the actions taken at any moment are not based on the latest input but based on predictions driven from deeper in the network uh from the same places that the sensory expectations are sent from uh this may be familiar to those uh who who are you know think like uh familiar with predictive processing or perceptual control theory and the idea here is that you know where the sensors and motors meet the network you know the the network itself is trying to make those local prediction errors zero meaning it's trying to anticipate the actual sensory experience um through kind of inverse modeling um so another just kind of quick look at uh you know how to look at this homeostatic process is through this process can be seen as generating an adaptive synchronization manifold between an agent and an environment um around minimization of prediction error um this is kind of looking at synchronization as prediction you know when when a system when two sentences are synchronized they can be thought of as predicting each other anticipating the other uh the others motions uh and individual nodes using ultra stability are using alter stability to learn how to predict their neighbors and when that equilibrium these nodes are kind of synchronized through these feedback loops um and the synchronization spreads um from individual nodes to whole network synchronization and ultimately uh out from the motors and backing through the sensors um in a synchronization manifold between the agent and environment um you know to say it another way the overlapping feedback loops interact and synchronize with each other as part of this homeostatic self maintenance and as these feedback loops extend from motors to the environment the sensors the model gains ability to anticipate the consequences of its own actions uh and the capacity to synchronize internal states with something in the external world the synchronization means that you know internal states will come to resemble uh the in one will come to resemble the internal states of the other um this is kind of along the lines of you know the good regulator theorem um when at this you know synchronized homeostatic state it's also in this critical state i mentioned this earlier um where it's stable uh but it's sensitive to local prediction error so the meaning that the the model uh this critical state is maintained as long as as long as local prediction area is maintained within a certain threshold um and but otherwise you know the smallest prediction error can trigger global kind

of structural changes within these feedback loops uh in order to better predict the future uh you know in order to maintain a synchronization network must be able to converge to equilibrium faster than the environment can destabilize it um and so there's um you know so there's update you know network update rates where we have to maintain um you know this kind of fast movement and this kind of is in line with the law of requisite variety where we have to make sure that we can explore state spaces faster than the environment we're attempting to control and as long as the homeostatic network can re-establish these stable feedback loops faster than then this synchronization manifold will be maintained uh and synchronization is kind of an interesting thing um especially in the context of um robotic control um because you know once kind of two elements are synchronized you know let's say you have a sub network that's synchronous synchronized then you know two indirectly two nodes that are not directly connected uh can end up coordinating without any kind of communication between the two because they've been previously synchronized uh their behavior will end up being correlated as well um and in case it's uh interesting to think about um you know what is the structure of these kind of feedback loops and this you know generative model and the way i like to think about it is a dynamic holarchy meaning um you know a hierarchy is kind of like uh you know a top-down construction um but you know a dynamic holocaust is more like a um a bottom-up construction where the parents are composed of the children and dynamic in that as the children change their own self-definitions the parents are likewise changed so it's kind of constantly shifting and changing over time as the children redefine themselves uh and this structure is very useful uh as a gender model as allows a small number of individuals to you know kind of reshape the hierarchy very quickly as it's part of an ongoing dynamic process um cool so i'll switch over and do some quick kind of uh demonstrations showing how this ends up you know why would you actually want to do this um you know yeah why would you actually want to do this especially in the context of robotics um so the first thing i'm going to show is um let me just share here quickly this is the demo i put together um a couple years ago actually let me let me shrink the screen a little bit so it's not quite so much to share uh this was put together a couple years ago to kind of demonstrate you know why would you do why would you do robotics like this you know what is the point um and you know what does robustness mean and what does additivity mean uh in the context of uh robotics so hopefully you can see now on my screen this is just a simple um modification of an open ai gym called uh reacher uh and what it's been modified to add a single texture joint here uh the reasons will be obvious here in a minute um but um this is just

a simple task let's try the you know the model has to get its end effector to the target position and this is using these homeostatic models here um but here's what's interesting i can do and it hasn't this model hasn't seen this situation yet i can disable the outer motor for example and it will basically on the fly figure out how to achieve the goal um you know just by updating its internal model and saying oh this motor doesn't work anymore let me let me modify my my model of the world i can turn that back on and um he'll start using it immediately or it will start using immediately i can do the same with the internal motor it can turn that off and so oops i did not turn it off there we go now now it's off um yeah so and it just kind of figures out oh okay the the inner motor i have no longer have access to let me update my model and i can again turn that back on and it'll start using it again um so this was just kind of a simple demonstration i made a couple years ago to show why you would want to do robotics this way in that you can kind of get this real-time adaptation to unexpected situations um due to the the generative model that's constructed this way um now let me just scoot forward a little bit to more recent work and hopefully um the frame rate won't be too bad over zoom let me just shrink the window a little bit in hopes that helps okay let me um share this the frame rate just feels a little rough over over a zoom here um but this is all just running locally on machine so this is something a little more uh recent a little more complicated we made the goal a little bit more complica complex uh and we've used a full six degree of freedom arm here from like an uh this is actually a fetch robot mobile arm um but something a little more realistic than like a toy problem uh and i'm just you know continue periodically here re-targeting the the desired angle for this valve and so this is just a robot that turns valves uh and what i've done is that um i i can periodically change the set point for its des you know where how it sees the world basically use defined goals by you know defining preferred states of the world uh and then the homeostatic model basically figures out how to control the robot to to uh to affect the you know the state of the world that it wants um i'm hoping here every 30 seconds it will randomize the um the valve shape the square one here is the simplest one but let me add some random motion and random orientation the model has not seen this before either so it just kind of learns on the fly how to deal with this kind of moving shape as i do it i can also do it with a sinusoidal motion here so imagine it's on a floating platform um where the rotation and the uh position are kind of moving through space um here we go now we got something more interesting so now we have it still moving through space it's a circular valve here something a little harder to turn um and what i'm going to do is i'm going to make it very slippery so imagine i've just poured oil over it so if now it's having to deal with a really really slippery

valve here that it has to turn and still manages to do it even though it's really hard um but you can get the idea here is you know uh if you do the generative model in this kind of sample efficient way you can kind of um get natural um you know adaptation to whatever it's kind of experiencing in the moment here um i think i've kind of i think this is maybe maybe i'll do the random motion again maybe that's interesting um but yeah there's all sorts of stuff you can kind of throw at this and go kind of deal with it i apologize for the frame right i'm going to blame zoom for that cool all right good um i don't think i have too much left you know next steps are uh basically improving automation and tools um i'm working uh my startup is basically developing this for um for real world use uh in adaptive situations where you need um fine motor control uh and kind of uh dexterous object manipulation in uncertain environments um you know it's on the technical side we're looking at you know scaling up both specifications and topologies which are related as well as you know deeper temporal horizons in both directions planning uh and ahead and recall um but yeah that's um about all i had for today uh and i look forward to the questions during the uh the um round table in a little bit if anybody has questions now i can talk to it as well i'm a little i went a little fast i apologize i'm a fast talker it was great just maybe one preliminary question what settings do you see these uh kinds of implementations occurring in first and then the other question in order or if it's your preference is how do you see the homeostatic network related to the analytical first principles foundations of active inference as we know it today yeah no um those are really good questions and i you know i i didn't talk much about active members because i figured everybody else had kind of done it pretty fairly um the use cases first your first question um use kids we're looking at are things that are just beyond um the ability of uh current robotics um the idea here is this software could be put on you know any old robot and kind of provide you know um adaptive capabilities for certain things um you know we're pretty open-ended in terms of what we're looking at in terms of use cases uh right now our current uses are in like um uh energy uh and manufacturing uh particularly um this is a fair amount of stuff that still has to be done by humans that could be pretty easily uh moved into robots um if they had you know a little bit more capability uh and then in terms of how does uh the homeostatic networks kind of fit into the active inference uh framework in general and i kind of touched on it a little bit in terms of how this uh these ultra stable nodes work in the framework of homeostasis uh if you kind of think of homeostasis as the network trying to seek you know global equilibrium um what this means from the ultrastable nodes point of view it means that it's um it's uh its local output state is is zero and that local output state is um the

result of an accumulation process of this linear transformation so the idea here is this ultra stable node what it's doing is it's doing a search it's doing a search through linear what are the right linear transformations that make my local uh state go to zero uh and so in that context you can kind of think of every time the um the ultra stable node updates its local linear transformations um this is updating its local model of the world and making a new guess about how the real world works uh and whether or not that's useful for the purpose of global prediction you know it depends on whether or not it makes that local uh value go to zero so there's there's a lot of threads here in terms of um expectations and um and you know prediction error flowing through the network um but it's a much more distributed process like it's and it's much less clear like um what each node is is what part of the model each node has taken place because you know each node in the network uh when it's changing its linear transformation what it's actually doing is slightly redefining all the feedback loops in all the of all the sub networks in the network it's in all the loops that flow through that node it's making a small change to so um each node in a certain way then has effect on all the other networks uh that are connected through its feedback loops and all those other nodes also have impact on its uh signals that are flowing through it uh and this is this reciprocal process that kind of builds this generative model i'm not sure if that's particularly helpful yeah that's great um if i could make one more comment question then we'll bring adam saffron on in just a minute or two the letter that you showed was quite interesting i got the sense of this like multi-generational titanic clash between what you described as infinite computation in the discrete setting with the turing tape and then infinite adaptation in the continuous setting with the generalized cybernetics and so i think that's a very fascinating framing because what do people say about computers and continuous time systems we're going to have to discretize and so the implementation is stepwise and that's what allows us to like actually implement it on hard drives and cpus and so on with the von neumann architecture but then also in the background or like even the water that we swim in is this continuous time the real time unfolding of perception cognition action outside the sandwich model outside of the discrete time yeah i i i'm really glad to pick up on that i wish i was more qualified to speak on such topics i'm a computer scientist so i i don't want to speak too far outside my field um but yeah i think you hit the nail on the head of my gut instinct on it you know um all right we'll take just a short break and then come right back with adam saffron all right welcome back everyone we are continuing on with adam saffron generalized simultaneous localization and mapping gslam as a unification framework for natural and artificial intelligences so adam thanks a lot for joining

and please take it away um thank you daniel so hello everyone i'm adam saffron i am a research fellow at the johns hopkins university school of medicine at their center for psychedelic consciousness research and today i will be discussing a grand unified theory another one but before i do um as was previously mentioned and as is probably of interest to people watching this i recently co-organized a special issue of frontiers in neuro robotics with inescipito and andy clark most of the submissions are already in for you to read and that will be closing up soon there's also another special issue that i am co-organizing with other active inferencers and fristonians for royal society interface focus on the subject of symmetries in mind and life broadly construed ranging from symmetry breaking and dynamical systems a gauge theoretic formulation of the free energy principle this is actively soliciting contributions and so please contact me if you have anything along those lines uh in this work and throughout all of it my journey i've been trying to understand the basis of autonomy and biological and artificial systems um do we have free will how does that work could we build autonomous agents that work like we do and have our capacities these are the kinds of questions that have motivated me towards this end i have been working on some fairly ambitious projects such as a theory of consciousness that tries to bring together various theories within the free energy principle and active inference framework and also models of free will in terms of the micro mechanics of agency across all of this work i take what i've called a marian neurophenomenological approach or a neil seth calls it computational neurophenomenology which i think is a better name uh and roughly the idea is you take the core aspects of experience um seriously as fundamental things to explain and then you cross-reference this with a multi-level uh functionalist handling or a marion handling where you can simultaneously analyze the system on computational or functional levels with the system um what the different aspects are for their their adaptive significance what their the function they're serving this algorithmic level the abstract way the specific programs and operations by which this is achieved and the implementational level or the physical realization of these and so the idea is you take this multi-stack under um understanding of cognitive systems and you cross reference this with um phenomenology taking experience seriously and this is my general approach and uh active inference um with its associated process theories basically checks boxes across all those levels you have the free energy principle you have active inference and you have predictive coding and these various claims of different degrees of specificity really help to flush out this mutually constraining multi-level account of cognition more recently i have been collaborating with roboticists and here are two particularly excellent

bio-bots tim verbalin and ozin kathal and um a lot of these journeys uh we encountered each other and um there's increasing interest in uh machine learning and artificial intelligence and things like artificial consciousness you'll have people like bengio talking about system two cognition and more um recently even lacuna's been starting to take these things on uh with um his jepa architecture talking about world models of different kinds and how they might serve the functions of artificial um but for my collaboration with tim and ozin we've mostly been focused on the specific problem of navigation um specifically a problem known as uh roughly the idea here is that um where am i in space and what kind of space is this this is a fundamental question for any active inference system any cybernetic system um but there's a kind of a problem here and that you're trying to map out the space and situate yourself in it in the same time simultaneously so but to map out the space it helps to know where you are in the space but to know where you are in the space it helps to help to have a good map of that space and so there's various proposals for different ways of bootstrapping yourself to degrees of certainty of what kind of maps or models are adequate for situating you in the world in a particular location and this is going to be important whether you're a robot navigating through the world or an animal foraging for value you got to know where you are to know what to do more recently in this collaboration which i've really found to be amazing because uh i just i'm in awe of robotics and what they can do it's kind of like that feynman quote of what i cannot build i would i cannot build i do not understand well they live that that's just what they do all day long they're like introspecting how do i work they're going into theory and then they're actually building a system in this back and forth iterative process and so the collaboration here was i'm looking at what ways can um biological systems and specifically focusing on the hippocampal system what ways can these inform navigation problems um in this more recent paper that i'm going to be focusing on more today called generalized simultaneous localization mapping or gslam we're reversing the flow a bit and the idea is more than cognitive neuroscience and forming robotics it's looking more to robotics to inform cognitive neuroscience basically leveraging the the precision they have with characterizing these problems and the way they're in touch with the ground truth of engineering and physical moving systems um using this to inform cognitive neuroscience and specifically the claim is that this slam problem is fundamental and that it may have structured a uh surprisingly maybe shockingly broad range of cognitive processes and that this might be a fundamental a source of fundamental principles for understanding how high-level cognition was achieved with high-level put in some scare quotes but the idea is the things that humans seem to be or animals

seem to be particularly capable of doing relative artificial systems and humans in some ways even more so and the properties we want to recapitulate in artificial intelligence and machine learning uh so some precedent for this um came from work with ratslam a seminal work by um milford wyatt and praiser so i'm in this they uh look to the the rat in this hippocampus and look at look at its wayfinding and navigation abilities and try to um basically reverse engineer these uh there's a really um excellent youtube video that i'll post um i've watched it countless times which just summarizes this process of building up rat slam from like these initial foundations of the ambition and folding in more and more details finding you know this isn't quite working going back to the biology going back to engineering and then actually showing systems that um do the same things as biological systems inspired by those systems it's extremely compelling um it kind of like you know every time i watch that video i feel like i get something new i wonder is are these the steps that nature took and evolving these systems i also kind of felt like what have i been doing with my life all this time like why didn't i find roboticists sooner and started talking to them um so i'm very grateful to be part of this symposium leighton and very grateful for uh tim and ozin with their work with leighton slam uh what they're doing is they're tackling the slam problem using active inference i'm not going to go into all the details here but some of this was discussed um there's parallels with some of the architectures they were discussed previously and tim touched on this so you basically have an agent moving through the world that has a variety of views of what it's expecting to see and poses and this experience map is a trajectory of the agent space and time and then here would be a generative model description in the middle of this where you have these conjunctions of posing of poses and views which are unfolding um in this series of transitions um adaptively sculpted sculpted via your policies but where there's this upper level which is basically organizing this as spatial temporal trajectories situating you in some kind of map or graph of space this conjunction of pose and view um one of the places i found to be compelling is that um i've arrived at these as basically the primary modalities i focus on just from like looking at the systems neuroscience but this um happens to be the primary things that are focused on for helping a robot to move through the world and a pose and view um and basically for neuroscience i focus a lot on systems such as the pecunius as potentially um enabling something like a mind's eye and the lateral prior courtesies of something like a live body and that these would couple very closely and that would make some sense since your pose would inform your view and vice versa but there's another detail you need for this to be an autonomous system an adaptive agent which is um not just like what's where is my body and

what am i seeing but where am i in space where do i want to go relative to the world and the various things that i believe are in the world how do i pursue my goals in the world and so enter uh the slam problem um but what's built with latent slam this is a you know this is a very impressive architecture um some details were described below which you know all these things governed by a singular objective function where the agent is able to not just um uh make its way through the world but also it's this powerful self-supervised learning curriculum as you're moving the two views views and pose and landmarks each one providing a source of predictions for the other training up the other um helping you to uh get a grip on the world in terms of slam um two of the things that um so this like broader notion of g of g slam or generalized language this would be a way of thinking of high level cognition um i'm particularly interested in uh these two operations one's called loop closure and the other is graph relaxation no duplication i guess that's three but the idea is um as to mention when an agent finds itself in a situation where its pose and view are highly similar to something to what was previously encountered uh this affords a potentially unique opportunity you now think you know where you are in space and so this is a really good opportunity to update your experience map and so coming back around it's not just a good opportunity in terms of you have high confidence but it creates a closed system so you closed a loop and so you basically have these different equilibrium points in this graph um which can be thought of as a map also but basically because they're all chained together um these are sometimes modeled as like energetic spring systems and they're all mutually informing and constraining one another and so when you find yourself with a loop closure event where you you're in this highly familiar state you then take this as an opportunity to refine your map and so you you know uh will do a very fine if if you're within a certain margin of error of um a location then if you think you're you really got it you'll then take the overall map and re and relax and you'll basically move around the various dependencies of different nodes um in space the different landmarks to adjust them and have each one refine the other and form the other however if you're past some threshold for similarity and it looks like you're close but not quite there you're a little bit different if you're surprised then you'll duplicate the node and then you'll create a new landmark and so now you'll have a new graph and so this is basically a kind of structure learning and there's a lot of work in bayesian cognitive science on um and within active inference a lot of open work like how do we effectively do structure learning for handling complex domains and i'm suggesting and uh i think timothy um hopefully would agree especially so writing together uh would be that um this provides a principled basis for doing

uh structural learning uh what's interesting here though um in this diagram that tim made is that as you're mapping out this warehouse if you set these thresholds for uh identifying a node differently if if your graph relaxation no duplication thresholds are set differently you get very different models or maps you might get very different structures or if you will schemas or scripts uh that's not the frame in terms of a slam problem but the claim here is that basically these the hippocampal uh supposed place fields provide a basis for structure representations many of which have resemblances the kinds of things that were hoped for in like good old-fashioned cognitive science and a neurosymbolic ai um and so some of and and part of what's hoped for is some of the um aspects of potentially uniquely human cognition high-level cognition high level are require some sort of representational scheme but if you set these parent but if this is the mean by which you arrive at representations and if these parameters set differently could give you um wildly different representations these would have very different impacts on the kinds of minds you might have potentially including even things like cognitive spectrums if you were to apply this to a biological domain and so there's some notion that for instance you could even potentially think of these parameters for the hippocampal system um which are maybe evolutionarily were originally laid down as parameters for this for a slam problem for an animal finding itself in the world but where the continued operate in terms of um the mechanisms of hippocampal and internal plasticity these if they're parameterized differently in different people could give you very different kinds of minds and this potentially has very um the significance of this might be difficult to overstate in terms of characterizing human difference these might be some of the most important variables you could look at for saying how people differ in the way they relate to knowledge in the world and not just so this would have clinical consequences basic science consequences maybe even helping to explain some of uniquely human cognition the kinds of things that help to get um symbolic uh reasoning and like cumulative culture off the ground there's other proposals that are kind of similar to some mad dog navigation perspective or slam perspective uh jeff hawkins out of numenta has a proposal he calls a thousand brains theory technically it should be called something more like a hundred thousand brains theory um where the idea is that there's a common cortical algorithm carried out by every single cortical column of doing this um object modeling and navigation of the kindness described with the kaplanatorinal system i i find aspects of this to be compelling but also very mixed feelings and i can talk about that later with anyone who's interested uh there's another recent proposal in this spirit by chris fields and michael levin where they're basically characterizing morphogenesis as cells engaging in a kind

of way finding a navigation process of where they ought to be in this um synchronization manifold of the phenotype and so they're all finding their place you know singing as karma might say like singing from the common hem sheet knowing where they are but as as they're synchronizing they're moving around to find where they are this uh unfolding geometric melody that eventually becomes the phenotype of the organism but the idea is that this intelligence of wayfinding at the cellular level many of these mechanisms may have been repurposed and recycled and deployed differently for overt navigation and that this is a frame for intelligence more generally i found this to be extremely compelling although what i'm doing with this work is i'm focusing more on the hippocampal enter rhino system as a source of specific adaptations with specific properties and a major transition unit and evolution potentially also of some connection to things like um ginsberg and jablanca's major transition markers and unlimited associated learning um they have a lot of emphasis on hippocampal system and its homologs there's very interesting work that i'm only beginning to look into in the insect and insects with the central complex and the mushroom bodies and the way they handle navigation i i think cross-referencing the mammalian version and the insect version would be really invaluable and that's like that's a a major to do that i'm hoping daniel will help me with some other precedence would be in terms of this idea of generalized navigation that this would be a powerful frame for understanding cognition uh comes from some older work with hasabis and maguire they found interestingly that hippocampal patients not only are they impaired in episodic memory but they have trouble with episodic imaginings of counterfactuals of novel situations and so they talked about this construction system of the spring this core system for memory and imagining allowing for various forms of um sophisticated modeling and creative cognition and this seems to have been part of what basically gave demis the the confidence to go and found deep mind and this is still a core aspect of what they're doing there and and the race is on for reverse engineering the system and the race is heating up and some really excellent work would come from uh timothy barron's group um i believe he's still at ucl um they're showing like for instance uh there's an architecture called the tolmán eichenbaum machine which i strongly recommend checking out um that's uh active inference compliant and as a realization of the functioning type of campus system they don't focus as much on navigation although it can be used for that and that's part of it that i'll come back in a second for the differences here but um when in this work one thing of note is so they're showing basically people categorizing these stimuli um where you have these birds with different length necks and there's a kind of morpho space in terms of characterizing the the

features and the various dimensions and basically they're showing hippocampal slash end to rhino-like representations grid-like representations being involved in helping to map out this feature space and so this is one example that's very commonly cited of spatialized cognition or something that might not seem clearly spatial but then the idea of gsum would be once we've spatialized it then to what degree in our sense making is it actually kind of navigation through the space and so this process of spatializing and navigating through the space simultaneously i would call that a slam problem and i think a lot of cognition can be fruitfully framed in this way and i'm going to go a little further so here's the tolmán i can bomb machine and he's showing like different um i don't know is his face covered by the thing i can't tell uh well there he is okay yeah so his work is unbelievable um his group and um but you know different things like uh relations among people within uh hierarchical um tree structures of their relations uh doing things like transitive inference of one property related to another and a set of dependencies but basically using the hippocampal systems properties to navigate navigate and manipulate these spatialized representations um there's other work from deep mind um there's this other talk highly recommended by um stachenfeld at all at the main conference it's unbelievable talk uh talking about the the recent work from deepmind to reverse engineered hippocampal systems uh place cells and the grid cells of the entorhinal system as allowing for um a variety of properties some of very similar to baron's work and they collaborate together but one thing that's interesting about this talk is um she makes a point that this isn't just um for tasks that are like physical space this isn't necessarily spatial um i might uh challenge that a bit um and and say that um we should basically always have a prior of looking for the spatial properties of any task domain and then wondering to what degree can we frame this as a slam problem and i think there's maybe two reasons of doing for doing this one uh maybe three uh one is that um basically putting things into a common framework helps to you to draw analogies across um seemingly heterogeneous um domains and this can be a source of insight in terms of a common representational framework for sense meaning sense making gives you an ability to have creative combinations from seemingly different domains so but the other reason is um the hippocampal system like this was specifically probably what it was selected for this is a mapping system for a animal foraging through value through various spaces and this isn't just evolutionary but this is also developmentally and ongoing this is an ongoing problem we have not transcended this we are still in space we still constantly have to move through space and situate ourselves through spaces physical spaces and also conceptual spaces that's part of the reason i'm really emphasizing the slam framing because hippocampal system

will never lose this job description and this the idea is that understanding the ways in which is parameterized specific operas operationalizations i don't think that's a word it's essential to keep in mind this ecological significance and so this is all within the spirit of um i guess a broader view in cognitive science called ecological rationality which is go back to the origins and try to understand it's tim bergen's quote nothing in biology makes sense except of the light of evolution take that very seriously and run with it one other thing that's uh so alongside one thing i can get is connections one thing a g slam perspective gets you is connections to forging theory and so um you know if you're thinking of an animal going through the world trying to extract value resources are often patchy and so as you're moving through the world you uh might be you know balancing expiration and exploitation with respect if you find a patch might be good to exploit that for a bit if it's particularly juicy but then as that patch dries up maybe it's time to go explore and then you want to move between adaptively move between phases of exploration exploitation staying longer in some places and then moving to other places and so these um jumps other places like when a patch starts to dry up and these are sometimes known as levee flights and this is playing actually into some of my work in psychedelics and that um informed by some really intriguing suggestions from matt shine's group uh the idea is that one of the mechanisms impacted by psychedelics might be um potentially facilitating these kinds of levy flights where you can think of more creative cognition as involving more exploration more broad associations and there's all and forging theory uh you know it's it's extremely rich and well developed uh there's all sorts of different modes of foraging that could be thought of as different modes of cognition there's specific hypotheses and connections to formalisms like the marginal value theorem so you would have like a hypothesis like when do you leave well in the marginal value theorem that is you leave a pact or when you when you go and try to find the new you leave a patch when the average when your current estimated value drops below what you think the average rate of return should be so this then you know gives you um basically like you can now characterize people where are they relative to these different normative models um and you know this idea maybe has an even older vintage or maybe you know newer relative evolution but old relative to discourses in terms of the art of memory and the art of memory and the method of loci you know memory especially before we wrote everything down or when writing things out was expensive and even now to have something in your mind where you know it could be at hand where you can retrieve it to have like a well-curated set of of knowledge it's essential and so one of the best tricks that were discovered is you actively intentionally spatialize your

knowledge you actually create a physical place that you visualize and you move through and at each place you at each point in this space you've placed different things you want to remember and part of the reason this likely works so well like almost certainly is this is um you know the core of memory in the campus system is actually involving a process of it is a spatial mapper that's something i would argue uh there's connection to some other more cognitive science i'm not going to get into right now but like multi-dimensional scaling where you understand potentially complex domains by mapping them onto a more a simpler and potentially more cognitively navigable space and then i'm not going to get into this right now but um actually i will so real quick aside is with respect to multi-dimensional scaling one of the early things that got people excited for about it was it's potentially used for analogical reasoning or establishing similarities between domains it's a very important thing to know like how similar is something relative to something else like should these things be placed in the same category or um you know it's one way of doing this was saying what is the distance in this when you do multi-dimensional scaling and you place something in a feature space what is the distance between these between different the further apart the less similar they but there are other models of analogical reasoning and there's another race on to try to figure out how we do analogical reasoning and so some of these proposals um you know they would go on beyond something like um multi-dimensional scaling but they involve specific structure representations but then we have a question what's the ontological status of such things do they actually exist in the minds we do actually have representations or maybe these things are more like they're virtualized or inactively realized or implicitly i suspect they probably we probably um many of the the kinds of structures you know described in good old-fashioned cognitive science might not be biologically plausible but they might be surprisingly biological plausible if you actually have a system capable of adaptively constructing graphs based on um your ability to extract value from these graphs and situate things relative to other things and so it's possible that some of analogical reasoning and even causal reasoning could be unlocked by having a good means of um creating structured representations and one of the things that the capital system might seem to be able to do is create these graphs where the different place cells of the campus can be thought of as nodes and so if you duplicate a node you are now basically doing structure learning um you can you have you're complexifying this uh representation uh so this comes into my uh work in other ways such as um models of goal-oriented behavior i'm not gonna get into too many details right now but here would be the campus system and you see um based on work of people like reddish and also um like sam

gershman has a successor representation uh model but basically you'll see these predictive campuses a predictive map or what's being predicted is um where you'll see these sweeps of activity that will proceed in front of the animal before and then which will predict which choice we'll do what arm of a maze in my coda the idea is that the hippocampal system is the top of the cortical hierarchy and that any prediction errors that are not handled elsewhere end up get temporarily stored in this volatile uh memory system which is also encoding these trajectories or these series of equilibrium points of state transitions and that um basically you have you know prediction areas that are not handled elsewhere make their way there get temporarily stored and you can have um what's sometimes called semantic pointer architecture where basically as you're moving as you're stepping through these various equilibrium points you can then unpack this as estimates of system and world um in terms of operative policies and actions and what you expect at different points along these different trajectories and so here i'm focusing on moving through a physical space in this case making tea but this is i would argue this is also part of what we do when we navigate conceptual spaces that we are um deciding what branches do we go down to what degree where will we find what kinds of value and that this is a similar operation using similar procedures from common neural processes more recently i've been trying to extend this to models of volition including um free will let's see i don't uh daniel how much time do i have left i'm sorry ten minutes okay great this is uh less i guess explicitly slam framed um but still it's a hippocampus at the center of it and so so here i'm trying to this is a recent piece that's in pre-print form which i would have uh i'd love feedback on but basically the idea is as you're that the present moment of experience uh has a thickness to it this is sometimes called the species present or the remembered present and that it seems to vary that the minimal thickness of a moment of sense making seems to be about a third of a second and the upper bound seems to be about three seconds and that this seems to correspond to um an epic so so you basically can so the hippocampus system can whenever you enter like a room it'll tile that domain with a certain degree of granularity it'll create these hexagonal tilings of that space and then within there that's where you'll have these different attracting attractors these different trajectories play out orchestrating the rest of the brain in terms of its dynamics to pursue these various goals and so the runway you have to work with in terms of the the thickness of the different rollouts for sophisticated inference would depend on partially uh the shelf life of this set of attractors before you have to refresh them uh in some way either set down an entirely new set or um uh try to recreate the old one or prevent the remapping and this would be related to some of what was what was just

described with aerodynamics in terms of how good of a grip do i have so let's say you have like a poor grip on what's happening and your prediction error is going up this is not a good situation for you that might be a good opportunity to stop hold and catch fire re-tile space get a new set of operative policies for sophisticated inference so here what i'm trying to do is show um different levels of uh agonism of these serotonergic receptors involved with psychedelics um five hc2a receptors are influencing the stability of these sets of attractors vividness with which you were doing these imaginings and of rollouts of actions into the future here it's it seems that what you could potentially do is describe some of a dose response curve for something like psychedelics where as the dose increases the extent of the rollouts become greater you've stabilized them and imagination become can become more vivid and can influence your policy selection more because you have more confidence in what you're seeing and so sophisticated inference your imaginings are contributing more to your ongoing overt action selection up to a point but then and so this would be and so this but then at a certain point um the idea is you keep dosing uh the animal or maybe the robot and it enters more of a creative dreamlike regime where it starts to lose coherence and you see the the extent of these rollouts of policy selection decreasing uh things are becoming more vivid but less coherent and sometimes you're getting uh and this itself could occasion error and potentially re-gripping in this model and so the idea that is that this would be a cognitive spectrum potentially in addition to explaining some of the neurophenomenology of things like psychedelics this could potentially explain some cognitive spectrums whether we're talking about maturation or aging or waking up you know when you first wake up to going to sleep these different parameterizations of your mechanisms for sophisticated inference helping to basically provide a neurophenomological handling of human uh of what it's like to be a person and all the variations of it i think that should be it for right now and uh happy to talk about all this with anyone who's interested thank you awesome thank you adam hi daniel hey and you're muted thank you very much any other uh notes you'd like to add i think that's uh i think that's it for the moment great hope that you uh consider anything and join us for the round table shortly so thanks and we'll be right back with the final presentation by jf clutcher all right we are back this will be the final presentation of this interval this is jf klutcher active inference for lego robots so looking forward to hearing this and then seeing you on the roundtable jf i'm jean-francois and i will talk about how i program lego robots as active inference agents this work contrasts with most robotic implementations of active inference which are based on probabilistic inference so i hope this will spark some interesting discussions well first of all

i'm a software developer i'm not a neuroscientist in my spare time well i like to code models of cognition because i'm keenly interested in all matters cognition and and i run them on robots because the real world is hard and i believe that writing code for robots is actually a very good way to test my understanding coding itself helps make abstract concepts tangible to me so what do i want to accomplish well i want my robots to learn autonomously while they conform to the principles of active inference this is the current version of my robots there are two earlier versions and there's one in early development and here we'll see my uh robots the left car to the right nd uh driven by the interplay of generative models for hunger danger avoidance and wanderlust let's see them in action they navigate his face the sheet of paper on the floor represents food and they are attracted by the scent of food simulated by an infrared beacon on a pedestal so cars making a beeline to the food while andy is um having a problem getting traction and bumping into uh the walls carl is cautiously approaching but we'll get a little bit too close to the pedestal and this will engage its danger avoidance collision avoidance derivative of models he backs out andy is observing all this and he backs up as well carl is reapproaching food and this time we'll eat successfully and he has noticed that carl was freaking out and in sympathy freaks out himself carl is eating all right so that was um the robots as they are right now after three iterations of this project so what it take what does it take to implement an active inference agent well there are three approaches one um let me take the numerical approach which basically puts the mathematics of active inference into code and runs uh that on robots um another approach is to take a symbolic approach where we have a code functions that operate mostly on symbols predicates etc here we can see we have the symbols for hungry very danger curiosity etc um but that's not the only option take e coli e coli is an active inference agent and its implementation is biochemical well i clearly didn't take that route i went this symbolic route well mostly because it's the familiar one to me and it provides me with a better scaffolding for my intuitions and exploration i also find that it handles complexity quite well when i started this project in 2017 my first thoughts uh were to implement a society of mind i'd come across this concept uh decades ago it's a theory of mind put forward by marvin minsky 50 years ago at its core it says that intelligent behavior emerges from a lot of simple actors that interact in simple ways it so happens that i look at it as an architectural principle as well so how does one implement a society of mind um to me the answer is clear is we use the actor model the actor model is a model of concurrent computation that was introduced by carl ewit in 1973. according to this model of computation independent actors which are processes concurrent processes each one managing its own state

and the only way a an actor can interact or influence another actor is by sending it messages and once an actor receives a message it will interpret that message possibly modify its internal state and as a consequence of fire messages to other actors so building a society of mind using the actor model is quite a natural fit i also happen to program in the elixir language and elixir is a message-oriented multi-process language which is based on the actor model so it's a it's a perfect fit so here we are it's 2017. i own two um lego mindstorms ev3 kits they're wonderful robotic skids with a a nice choice of sensors mini motors and computer the ev3 brick that controls them all and i want to build a society of mind in elixir on my eb3 robots i'm in luck there's a linux distribution called eb3dev that allows me to run elixir on the ev3 brick so i'm all set that start that allowed me to start this project and this project as i said has completed three iterations and i'm starting a fourth one and we look at each one in turn reviewing the goals and also the issues uncovered first iteration 2017 version one a simple society of mind as i call it i implement a an ad hoc model cognition it's not something that i i took from research is something i i made up myself it's homebrewed and let's have a quick tour so being a society of mine is bucketed with uh and um we have the detector actors and their responsibilities are to interface with the sensors and to pull them periodically to generate percepts which are events that are like all other events that the actors will produce are sent to the central nervous system actor which is a dispatcher which will then dispatch these events to other actors that are interested in them there's a memory actor and all events uh are sent to the memory as well to be stored for a certain amount of time and represent the past if you want to the agent and the other actors are able to query this memory in order to make whatever decisions they need to make one of the key actors are the preceptor and what they do is they ingest events perceptive ants originally from the detectors and analyze them maybe in the context of the past and produce higher level percepts for example a preceptor might be digesting a distance a percent from a detector look at it and say ah given prior distances i'm getting closer to an obstacle and in a higher level uh and then produce that percept and the high level preceptor might say ah i am getting closer to an obstacle and i am perfectly close to it so uh producer percent that says collision imminent and so forth and so on another important part of actor is the motivator and motivator is responsible for deciding what the robot needs what the robot wants and it expresses this need or this want as a motive event that is again centered in central nervous system dispatched in this case uh to the behavior actor and the behavior actor then takes this motive and acts on it by emitting intent events listened to by the actuator actors who translate these intents into actual commands and move the robot one key point is that the motivators

actually are compete with one another so that the most important need of the robot is the one that's expressed so for example being getting to safety is more important than uh um satisfying hunger there's um let's look a little bit into one of the types of actors the behavior actors and uh the behavior actor is actually quite complex it does a lot of work it's not a simple actor it's implemented as a state machine and with every state transition triggered by a new percepts intents are emitted and so we can have a a behavior actor which job is to uh direct the uh robot to the food so um it works so let's see nd version one moving about and trying to get the boot which in this case is the blue paper on the ground so though it tries to avoid collisions uh it not always succeed here it finds out it's stuck backs out goes around doing a better job at avoiding obstacles and now makes a beeline to the food having detected it approaches cautiously and feeds all right so this worked quite well but there were issues with this version one uh not this is i don't think it was visible in the video but the robot was frequently overwhelmed by a constant flow of percepts out of the receptor the detectors constantly feed percepts into the society of mind and after a while they kind of back up and the robot starts reacting to old events and one solution was to actually put the robot to sleep once in a while paralyze it let the events wash through the system and then wake it up with uh with a clean slate so that was kind of a little bit awkward and another issue was that well there's no learning going on the robot doesn't get better with each run and given the model it wasn't clear to me where i could fit uh learning in there so time for uh version two so i need to find a way to make my robots learn and they also need to um better focus and be able of attention so they can avoid being overwhelmed with all these unneeded sensations i also need more computing power so i replace the eb3 brick by a raspberry pi 3. about the same time i come across andy clark's book surfing uncertainty and i find that the predictive brain makes quite a lot of sense to me i think it tells a really good story about learning and attention so i decided to implement a predictive society of mind this is my first crack at it i won't go into uh the details i do in a companion paper um let's just say that they are actors for managing beliefs validating them making predictions and taking action when um to make these predictions come true there's also an attention actor for turning detectors on and off as needed and a focus actor for prioritizing belief validation i do introduce some learning the robot learns what actions tolerate correlate well with success but that's about it there's the the model does not feel quite right to me at this point and they are bottleneck actors for example there's only one experienced actor there's only one attention actor only one memory of course and only one focus in a society of mine that just doesn't feel right yep it still it works and let's watch andy um at on its first run

it's nd version two hasn't learned anything yet so the same scenario so andy has detected the food would like to move towards it but doesn't know which actions validate believed best it smells food but still you know it's just heading into the wall so for indeed it's been quite uh unproductive here in its quest for food it's a little bit painful to watch let's stop um now here's andy uh 30 runs later and it's much more competent about matching beliefs and actions smells food and very confidently makes a bee line to that food here we go slowly approaching obviously a very different anti from the the naive one and there we go okay so it worked uh but there's still uh there were issues with version two first the model doesn't quite feel right um to me it just doesn't quite capture um something didn't i didn't feel right about it and they were model bottleneck actors as well which is not a good sign and learning was limited to action selection so time for version three i call this one a society of generative so i go back to the andy clark's book and uh and also read on the free energy principle um and it becomes clear to me that generative models are actually quite central in fact it's generate models all the way down so i get to work on a new model in this new model i have very few kinds of actors they're detector actors as before to interface with the sensors actuator actors to interface with motor speakers and whatnot and the only other kind of actor are the generative model actors the gm actors and there's quite a few of them in this case here uh 16 and they do the heavy lifting they do the belief updating the predicting the raising prediction errors the selecting selection of actions and they are responsible for remembering their past states and and actions um each gm operates within its science scope there there's one for uh locating food one for avoiding obstacles one for a detecting approach to obstacles and so forth and so on the gm actors have parent-child relationships apparent gm sends predictions as events about what it expects its children gm's to believe even what it itself believes and these prediction errors these predictions flow downward prediction errors flow upward if a child gm receives a prediction about what it's supposed to believe or it's expected to believe and it's not what it believes and it sends back a prediction error up to its parents well that means that a uh generic model's perceptions are in fact the uncontested predictions it makes plus the prediction errors it receives and the gm updates its beliefs based on these perceptions and given what it believes it may decide to take some actions and then raise intent events that reach the actuators a little bit like before the generative models are kind of grouped uh into different areas of concern there was uh there's being which is responsible for deciding whether we're in danger we're hungry or free there are the danger uh gms which have to do with uh clearing obstacles uh trying to stay in well-lit areas avoiding collisions with other the other robot and whatnot then

there are hunger gms um that have to do with eating seeking food etc and there's the freedom gm which i call the wonderlust gm which basically says well i'm neither endangered nor hungry so let me just roam around and finally there are these two ones here which have to do with guessing the intent of the other robot and we'll go into more details with these two so let's look more closely at the gm actor remember that a gm as a as parent gms and child gms that the predictions flow downward and prediction errors flow upwards let's look inside a gm defines one or more conjectures um a conjecture is a potential belief for example whether i know where food is or whether i'm about to collide i have two different conjectural beliefs a belief may either be a goal that's a belief we want to make true like i'm eating or an opinion which is a belief that i want to validate for example that i am approaching food a conjecture is typically activated when the gm receives a prediction about the conjectured belief and a conjecture knows how to check its associated belief to see if it's valid it knows how to make predictions given that belief and how to decide what actions to take to either achieve or test the belief a gm learns how well alternate courses of actions policies correlate to achieving validating belief it also keeps track of which child gm it trusts more should they send competing prediction errors so it does precision weighing and each gm actor works in its own time it's a concurrent and it does it it operates one round at a time and we'll see what a round means and it remembers um it's fast rounds uh it's past perceptions the predictions is received in the past its past beliefs and actions so now let's look at the uh life cycle of a gm actor well as i said agm actor operates round after round after round so what's around first the window round is started gm carries over the perceptions and uh beliefs from the previous round it doesn't start from from scratch and if it's in the midst of trying to achieve a gold belief it may at us as it starts you around activate conjectures for that action belief and um as a result send predictions to child gms once we're done starting the round is active and then the uh gm actor just sits there for maybe i don't know 300 milliseconds waiting for events events like predictions coming from above from parents that will then activate uh conjectures trying to prove that the predicted belief is actually uh true this will cause predictions to be sent to child gms and as a consequence maybe prediction errors will come up and the gm will update its perceptions when the the round is completed either because we've heard from all the children or because time's up the gm completes this current round it updates uh position weighing keeps its trusted perceptions updates its beliefs uh as a result may send prediction errors because they could the new beliefs conflict with uh predictions that it had received uh before it updates its policies it decides on the course of action since uh intense and notifies the round

it's completed and we continue so round after round after round i mentioned that uh there were two very uh kind of more interesting gms in the society of uh generative models these are these two gm's and they represent uh my little foray into theory of mind these gms are dedicated to guessing what the other robot is doing so each robot observes the other and detects patterns of movements and infers intentions from these patterns there's only two kinds of intents that inferred one is that the other robot has detected food and is making a move towards the food or that the other robot is in danger and hispanic now how does a robot see the other robot in addition to the array of sensors i already had from previous version i've added a new sensor a 360 beacon seeker and what this beacon seeker sees is the beacon um on the back of the other robot so each robot has a beacon on its back that broadcast and then 360 radia radius circumference angle signifies here i am here i am here i am and the other robot can detect how far the other robot is and at which relative orientation that let's say one robot here uh carl is making a bee line to the food andy has been observing this and seeing the pattern of movement which is a very uh straight uh trajectory uh infers that the carl has detected food and is homing in on it and may decide to follow carl and get to the food as well another pattern of movements would be a chaotic pattern of movement and i'd say carl had been backing away in panic from hitting the pedestal and andy was observing this and it noticed that this thing kind of a chaotic pattern and inferred that a car was in a panic and decided to get into panic as well if you remember the video at the beginning of the presentation this is doing well we all know debugging is hard i find out also that debugging societies of mind on robots is very hard in order to help myself i develop a simulated simulation environment i think of it as simulated embodiment which allows me to observe what goes on inside a given it helps for debugging it helps for experimentation in the simulation word world a robot my robots virtual robots live in a grid world there are obstacles the blue tiles there are uh kind of darker areas that the robot tries to avoid which are represented in the black and gray uh squares and and then there's the food um which is the green uh tiles here as the robots uh navigate that space you know by using their uh society of generative models i can see where each robot is in terms of its state first i can see where it's located i can see its orientation when it's last said what the last action it executed and i can see the state of its virtual sensors and virtual actuators i can also peek inside the various generative models of a robot here i'm looking into the avoiding obstacle gm of nd and i can look at the state in the current round or in past rounds and i can see which predictions came in which conjectures are activated what are the current perceptions uh what beliefs that the robot uh

holds at the time and um what actions it thinks are uh are more likely to work and what actions it is taking very very helpful but this moves really fast and so i i can actually slow down the simulation even pause it and then look inside the various generative models and see if what's happening is what i expected would happen so create debugging debugging tool and experimentation tool so i'm quite happy with version three i think it aligns quite well with the active inference ontology um there are no bottleneck actors and but learning is still limited to selecting um action policies however i think that the version 3 sets the stage very well for the new iteration and this one has an ambitious learning agenda and it's just getting started so version 4 starting now i call it active inferencing so here's what i want to achieve as the robot interacts with its environment and i and learns how to maintain it i wanted to learn how i want it for i want for it to learn how to maintain its homeostasis by uh growing uh its own society of generative models so instead of being given an apiary set of generative models i want the robot to learn that set not only that but i want each gm to learn its own capabilities i wanted to infer its logic programs that when executed will infer predictions from perceptions will infer beliefs from uh also from its perceptions and infer policies that will validate or invalidate its belief so how does one do that well there's a very very interesting paper that i found which is uh called making sense of sensory input written by richard evans and all from a deep mind it answers the question of how do you uh learn um a causal theory explains a sequence of sensations and once you've built that causal theory you can then apply it to predict what the next round of sensations or perceptions will be and thus make predictions about them now just to to read from the beginning of the paper making sense involves constructing a symbolic causal theory that both explains the sensory sequence and also satisfies a set of unity conditions now unity conditions are actually a set of constraints derived from cons philosophy that ensure that all the pieces of the theory form a coherent whole semantically speaking i won't go into the details but it's quite fascinating the paper goes on making sense of sensory input is a type of program synthesis but it is unsupervised program synthesis so the the paper presents an app reception engine this is the software that will generate will infer causal theories from sensory data it is at its core it's a code generator what it does is it synthesizes logic programs that implement these causal theories and when you run these logic programs you can predict incoming sensations so essentially a perception engine generates uh predictors and the way it does that is it searches for predictor logic programs in a space of potential possible logic programs it's a search problem and it's it's the the search space is extremely large of course so we need a way to constrain it we need to apply sorry strong inductive biases

to restrict the search that's where the unity conditions come in in order to derive its own predictor a gm will search for good predictors using app perception so given around a prior round of perceptions remember that the uh the gm has a memory of each round and in each round a set of simultaneous perceptions given that sequence a gm will search a space of candidate predictors for a good one what makes a predictor well two things first there's a scope and then there's rules the scope is essentially the objects the rules will be about and the belief predicates that will be used in the rules the rules well there's three kind of rules there's the static rules which apply on a set of simultaneous perceptions like at time n uh in it the rules validate and possibly also infer uh implied perceptions missing perceptions an example of a static rule might be if the robot is at distance zero from an obstacle it is also touching the obstacle then there are causal rules and the causal rules essentially infer the perceptions at time n plus one given the perceptions at time n they predict the next perceptions an example of a causal rule would be if the robot is at distance x from an obstacle and the robot is approaching the obstacle in the next round of perception the distance of the robot to the obstacle should be smaller and finally there are conceptual rules which are rules that kind of make the whole predictor hang together semantically there are rules of mutual exclusion for example there can only be one distance between a robot and an obstacle and rules of uniqueness i'm sorry the rule of uniqueness would be that they they can only be one distance yes rule of uniqueness they can only be one distance for example between two objects and mutual exclusion might be um a robot cannot both approach an obstacle and be touching it at the same time so the app reception engine of uh will kick in for agm when the current predictor of the gm is no longer good enough for the jobs like we've we've produced a predictor and now we have more rounds of perceptions and the predictor is doing a really bad job at uh predicting the next perceptions and then the perception engine will kick in and try to produce a better predictor for that gm so that happens for all the gms and once we have a good predictor the gm even the current set of perceptions can infer the next set of perceptions thus make predictions so we talked about uh how a gm learns um but what about how it learns how about learning uh the entire society of gm's how does a robot form its society of gms through interactions with the world this was going to be guided by a uh a midtech cognition gm that's something that's given when the robot starts live it has a basic set of detectors set of actuators that is a given and it also has a metacognition gm it's also a given metacognition gm its role is to watch over and guide the evolution of a society of gms that's the action for it to function though each gm must act as a cogniti cognitive sensor it must produce sensations such that such as um are prediction errors

trending up or down uh do policies uh particularly uh predictably impact uh my beliefs or very importantly within my purview is the robot's homeostasis at risks these are cognitive sensations that a gm will produce will be uh absorb uh listened to by the that will allow it to update its own beliefs about how well the society of gms is doing uh does it have sufficient coverage is each gm capable enough and based on that um may take corrective actions by creating new gms possibly if one gm is not doing very well that is stuck in a rut um removing it and replacing it with another so that's the the role of the metacognition energy how can the robot tell whether it's doing better or worse than before given what it has learned well feelings that's how so a robot will have internal sensors for feelings and i'm thinking about feelings of hunger which increases uh with physical and mental exertion and is reduced with eating there's the feeling of pain which would go up with the number of collisions but would be reduced automatically with the passage of time simulating healing and then i'm thinking also of the feeling of only goes up is if a gm is not learning anything if it's logic programs are unchanged for a long period of time and but on me goes away when the the gm is learning feelings uh i have a balance positive negative i'm hungry very hungry negative balance i'm not hungry at all positive balance and a negative balance signals uh risks to homeostasis when uh beliefs are derived from um feelings with a negative balance or a positive balance then these beliefs take on this balance this is going to be used to uh prioritize actions because actions that promote beliefs with a strong balance will be prioritized for actions that do not so uh in other words a a gm with very strong feelings will have its actions prioritized over uh gm's that currently uh want to take action but from uh you know less less uh less feelings less powerful feelings and if these actions um uh conflict then the actions of the the gm with stronger feelings will uh predominate will take over well given all this what do i think makes a robot an active entrance robot well there's many things but i'd like to think that it's in its simplest expression for me at at least um an active uh robot is one that learns what to do in order to feel good uh more often than not so i'm hoping that this uh work will raise some uh big questions and and give me some insights into them uh for example what is the appraised knowledge that an agent must possess to learn autonomously what makes unsupervised learning converge on competency diverge and maybe a little bit more far out what can growing a society of gms learn from developmental biology i suspect that there are principles at work here that both that they both share so that's it thank you very much thank you to the active inference institute for their help and encouragement um the version three of uh my work is available on github it's open source i can be reached on discord and by email and the companion paper to this talk is

available on senator all right great talk jf we'll take a quick break and we'll be right back with the round this is the beginning of the second round table that's our last session in the second interval of the second applied active inference symposium and some more may be joining but i am joined here by adam saffron bruno lara and jakobs mecca maybe just to begin it would be awesome if anybody who wanted to do thank you and welcome to jf carl will be joining shortly um so perhaps to begin if anybody would like to provide a reflection on a talk that was not their own what was something that you felt like was brought to the table what was a key problem area that was addressed and how did you see active inference being implemented and adding meaning in that situation i can maybe uh quickly just comment on uh jf's presentation since was the most recent i think uh definitely this symbolic approach to active inference is one that is has perhaps not been that pushed forward in and like obviously there are different um approaches all around the field but i feel like that this particular perspective on pursuing this kind of emergent generative models which only have the very basic um kind of setup but then the complexity emerges through the usage of elixir and prologue i think was really interesting yeah i mean we talked about js project in the organizational meeting units quite a lot every time i think about it it's really cool to see this implementation no i mean in terms of the recent one just the app reception engine like the content inspiration and i found that to be really exciting um especially because you know uh i've been taking some inspiration from kant some of the work and like wondering whether basically like these categories like are they not just like preconditions for judgment but like any kind of sense making whatsoever like any sort of world with composition that could be navigated and parsed um to see that that kind of work was being done with that level of sophistication was just amazing it is a thread that came up multiple times with juxtaposing discrete time continuous time and all of these and then it's almost like another dialectic with the numerical and this the distributional approach variational inference sampling message passing a whole taxonomy of approaches that we've seen for active inference models and also models of perception cognition and control outside of active inference maybe jf if you wanted just to provide a thought on where you see the symbolic perspective coming from and intertwining with active inference directions and robotics like we were talking about today well i i think it comes in from different uh uh perspective ones is explainability of of uh choices and behaviors if uh the um the the robot is driven by a symbolic reasoning then it becomes relatively straightforward it to provide an explanation of its behavior in terms of the chain of reasoning that led to it so that's that's of interest uh in and of itself the other aspect is uh when it comes to the app perception engine it has a an ability to

develop a causal uh theory a symbolic causal theory from relatively small data set so unlike machine learning solutions which require intensive uh intensive learning process which leads to an obscure causal theory if you want uh with the appreciation engine you have a relatively small amount of data that that as input and a causal theory that's symbolically expressed as output which can serve as a basis for explanation and justification and rationale that being said uh if the um as part of the uh process of perception uh gm uh uh abduces latent objects and and induces uh predicates uh the meaning of these predicates are uh only uh to be inferred in context because they're gonna be called like p1 and p2 and object o1 object o2 you may by examine emanation just maybe infer ah i think that we're looking at a predicate that has to do with uh navigation and and and avoidance but that will not be clear so i i think uh in that regard there there are gains but they're not they're not absolutely clear in terms of but when it comes to active inference i think what matters to me is that the robots behave according to the principle of active inference and not necessarily by implementing the mathematics of it so i don't think it's a contradiction as far as i'm concerned these are four diseases that are our mind just symbolic programming and that's why i'm doing it that way thank you it's almost like in the textbook of active inference where we have the high and the low road and here's like the symbolic airdrop actually arriving at a similar place although being quite disjoint in its origins from the particular partition flows on continuous numerical variables a lot of the ways that we've seen things develop in the past several years welcome carl if you would like to say hello or provide any remarks on anything that you've seen at the symposium or any overview thoughts on active inference and robotics well i've greatly enjoyed all the presentations there's so many different perspectives so uh um i will i would imagine my comments will fall on from what people want to talk about so i'm going to just pick up on that last theme so you know that jean francis was um symbolic dropping on from high i i thought it was a really intriguing perspective um and it's a perspective that i've been actually forced into in certain applications of the uh golo fashioned mathematics of active inference so i thought it'd be useful to share that um in the sense that when you write down discrete state space models and use the maths of uh minimizing variational expected free energy to invert those models very often you're confronted with specifying those models with boolean logic you know if this then that um and um you know literally in the code which you write to specify a particular tensor that maps from some causes to some consequences you are literally using if and then statements and and or statements in order to generate the tensors or or or the um the matrices that sort of encode probabilistic mappings which now become deterministic mappings simply because you're

populating these tensors with zeros and zeros and ones so i do see an enormous opportunity here to actually um bring together the um you know the mathematics and the simple probability theory just using you know um categorical distributions or dirichlet distributions of a particular sort you know where you're only dealing with some zeros and one probabilities and the kind of logic and perception that you were talking about i thought that was absolutely fascinating and one one thing that intrigues me if you can do that then that there is um a way of doing your searches of all potential programs or logical um statements that are internally consistent um with the evidence at hand you know for example i could build a generative model where i can be in two places at the same time or two things can be in the same place at any one time that's you know from the point of the genitive model i can write that down without an exclusion principle um but when i now come to see who asks the question is that model relative to another model that can be written down in terms of boolean logic that precludes two things being in the same place any one time does that simpler model now provide a more a better explanation for the evidence at hand so technically what that means is that i can apply the rules of bayesian model to this model versus that model where the model just is a statement of propositions uh logical propositions um and there are ways of um there are ways of sort of automating that very efficiently called bayesian model reduction and i'm just wondering whether you could take a lot of pressure off the search the internally consistent logical structures that you're dealing with and if you can cast it in terms of in terms of bayesian model reduction which quite simply is measuring the evidence the probability of this sequence of outcomes that can itself be categorical you know i'm in this state all that state the probability of that particular sequence of discrete outcomes given this structure this generative model that is articulated as a logic and if you can do that i think that'd be very very powerful because that speaks to a common theme which i think most of us or you were talking about which is this issue of structure learning um you know and what you're talking about is sort of a principled way to get the right parsimonious structures that have this you know this coherence i can't remember you you were even taking it right back to sort of 19th century philosophy or 18th century philosophy in terms of in the constraints would be really great to see whether that kind of um sort of logical structure uh the philosophers like um actually could be discovered or disclosed just using uh good old-fashioned basic model selection that there seems to be a close symmology there with some of the work of people like josh tenenbaum where he's he's he certainly articulates structurally learning from the point of view of radical constructivism using sort of programmed searches and sort of um you know casting generative

models as programs and then it's trying to score the quality of these programs uh using a form of basic model selection and he claims to be able to do sort of millions um very very effectively with it within um you know within a few seconds have you have you looked at josh chellenbaum's work at all not yet but i certainly will and i just want to uh make clear that that the appreciation engine uh was developed by richard evans uh in the paper making sense of sensory input um and and yes one of the the the core issues is inductive bias uh how do you restrict the search space so that you can have a chance of finding within a reasonable amount of time a logic program that is a competent predictor and those and that's where cons synthetic unity of our perception uh plays in as it provides constraints on the the class of uh logic programs which are uh acceptable which are are yeah which are uh considered valid if if i make a contribution because this is work in progress i would like to add additional constraints of from the fact that we do not uh we're not trying to generate one generative model in in in and of itself but as in the context of a society of generative models so general will borrow its perception domain from the belief domain of other generative models thus creating additional constraints and as it searches it will try not to modify its scope in such a way that it orphans other generative models so as the society of general models grows so does the same set of constraints that apply on reducing the search for a predictor in any of the generative models themselves so i'm hoping that as the society of general models grows so does the set of constraints that apply on the searches for any one of them but i think the name of the game is very much in the richness and the strength of the inductive biases that are applied on search just to highlight a few pieces of what were addressed and then to bruno or adam or anyone who'd like to add it impressed on me robotics is radical constructivism not just in the sense of the constructing models but also in the way that adam described in terms of the relationship to the embodied artifact so that is one interesting piece and another in this quest to find the rainbow that connects first principles and analytic formulations through mesoscale for example computational code all the way to the last mile in the cases that we heard from one hot chen in the previous session carl when you described the way that boolean logic enters at the level of execution was very interesting and suggests that there is kind of like a uh a brackish area similar to that between discrete and continuous time models in like hierarchical modeling but the numerical and the statistical approaches to active becoming connected to actual logical implementations and design choices that have to do with which functions are executed first or the conditions that certain things occur and so there actually is a meeting ground or an interface in a way implicitly already and so it is very interesting to see how jf's

work on the generative model generation is now approaching that intersection from the other side and suggesting it's like if you see a car coming from the other direction there's a road and now this high and low road network starts to grow more pathways bruno or adam would be happy to hear any thoughts you have um yeah thanks um yeah i mean i think that's that's um that's much our case as i was telling you like we come from this other side of of the story and at the end we have to make some choices on implementations and so far we've been very focused on on these um self-organizing maps most of our work is on that uh and that's actually one one of the questions i have for for all of you like what's your take on on all these approaches that we're trying to like we're coming from a different field from a different point of view and in a way we are almost uh intentionally avoiding uh probabilistic approaches to active inference no i don't know what's what's your opinion on that that was the first one and the second one is more like a um i don't know like a a doubt a question for carl like what do you see in the near future on on all these things so that we're working on on prediction aerodynamics and this monitoring of the of the error over time can yeah i think that's that's i'd i'd love to speak to that if i can in fact interestingly it does actually also speak to your question daniel i just wanted to point out also that this notion of moving between continuous states-based models and discretized state-space models is something that people in quantum computing and physics are going to keep their close eye on because if you get all digital um then the opportunity for quantum computing raised it raised its head so it was also very pragmatic um uh pragmatic reason to sort of look at that um that transition and also just to bring in uh adam's point you know not only um was he suggesting that we quantize discretize spatialize in terms of little tiles and places uh receptive fields for being here as opposed to being over there but he's also suggesting we do that in time so i think the quantization of space time in a generative model is a really important thing of course you know you could argue that that's a step closer to good old-fashioned symbolic ai that has a meaning in in the sort of focus psychology sense so to come to bruno's question so the big divide here is between um from my point of view at least degenerative models that are of continuous states and time versus genetic models that are of discrete states and states and time and clearly you need both yeah in the sense the the world that a roboticist and indeed me and my children live in is a continuous world i have to move around it i have to um control continuous temperatures and all sorts of things and and yet it seems as if the the intelligence is at a symbolic level so i think what we're talking about is a hybrid genetic model and then the question is how do you get the sort of discrete parts to talk to the continuous parts but if we just focus on the

continuous parts um and just to answer bruno's question directly um first of all what i'm saying is you can't ignore the continuous bits so everything you would talk about in terms of self-organizing maps has to be there at some level um it's just a question of whether you can put a more symbolic um logical structure on top of it to cause screening sufficiently to resolve all the complexity um the i i i loved you your work um i could see uh all the important um issues in terms of um representing uncertainty as the if you like the stand-in for the valence of how i am doing so i was amused to hear that you're trying you're actively avoiding probability theory yet for me everything you said was all about probability it's all about uncertainty and that was at true at so many levels so just to answer danielle's question to you uh the relationship between free and variational free energy and the prediction error is really trivial and very simple it's just under continuous state space models with gaussian random fluctuations the uh free energy gradients are just are the prediction errors the signed prediction errors so it's really your simple quadratic forms for the log probabilities um so um self-organizing maps that organize themselves in terms of responding to prediction errors just are systems that are performing a great inflow on a variation free energy under the assumption you're dealing with continuous states that have gaussian by random by central limit theorem uh gaussian random fluctuations so i i you know that that's all exactly what you know what would please me if i was if i was a probabilistic committed to the probabilistic punishment you your i think your thing you're bringing to the table though is is this sort of the dynamics of the uncertainty or the amplitude the prediction error the unsigned prediction error the sort of the behavior of prediction errors over you know with the separation of time scales and i think that that that's a really important um really important move um i i see slight homologs of that not formally identical but certainly um homologs of that um um from the point of view of an engineer this would be like getting the kalman gain right you know if you interpret a calm and gain on a kalman filter as assigning the right precision or confidence to the prediction errors as they come in relative to to the state estimators and your your prior beliefs that you've accumulated getting that right having an adaptive kalman filter where you're actually optimizing the calm and gain uh that would be if you like the state of the art hierarchical camera bc filtering so that if i was a psychiatrist i would say this is exactly what goes wrong in people with uh schizophrenia and autism is that they've got their estimators of the overall uncertainty or prediction error wrong um and that they now need to um you know and this is an explanation for um false belief updating of false state estimation from the common filtering point of view the the other point um that struck me in in in the early part of your talk was um something

which um as a neurobiologist or motor control theorist um something that they would find very entertaining which was the link between the ability to ignore stuff and sensory attenuation i don't know if you've come across that in robotics but certainly in terms of motor control of the kind i think you're using which is which i would um ascribe to the equilibrium point hypothesis like approach approaching so you you're putting your set point into into the actuator and you're letting the reflexes do the rest but you've got to control carefully um how those prediction errors are used to drive the actuator make sure they don't come back up and change your state estimation and so from the point of view of sensory attenuation that just is attenuating the precision or the inverse variance or estimation of the overall amplitude so and finally you know looking at the long term trends in the estimated precision i think is you know it is a really important way to see whether you've got the right generative model for this context um and i noticed you were you're citing the work of um manchester united uh you know and that was i think he's really great insight many years ago interesting from economics he was working with economists at that time so so uh i think that's very very important as well it's all about estimating uncertainty yeah thanks a lot um i mean as you can see there is uh well obviously we come from this this side of wolpert and you you can you can see all all that literature behind our work and uh but yeah we're getting there and um that's that's uh that that's where we are going that's what we're intending and you'll see some some work that we're going to come out uh very soon that tries to to bring deeper this this this um to make more useful uh making better use of this of this um monitoring and the other thing i was telling you i was thinking is something you mentioned is this waiting of the attenuation and and that's always been in our in our mind again because of uh why can you tickle yourself and all these famous famous works um we have lots of doubts we have uh lots of discussion with with with people about this attenuation and and the precision waiting and that's what we are now trying to to implement again using self-organized maps but uh yeah we we we try to get results quite soon and thanks a lot yeah just just to say i think that's really important because as soon as you can get dynamic dynamic attenuation or gating um in play very much in the spirit of what was implicit in um matt's um sort of homeostatic architecture you can talk i think genuinely you can talk about your your robots attending to this or ignoring that in the right kind of way and you're starting i think to get quite close to now sentience because you've got attention in the mix you've got some a kind of metacognition in your self-organizing maps which i think license is something which is much much closer now to sort of a sentient intelligence as opposed to just a reflexive kind of control yeah that that's that's that's the idea let's let's hope

for the best a few points and then adam or jacob or anyone um active inference is inference about and it also uses action as one of the ways to reduce uncertainty and so it's a very tantalizing parallel with quantum active inference as being about quantum systems and potentially using quantum in some way to run these models another point was in the recent folk psychology paper with with ryan smith at there was a distinction of how increasingly cognitive and symbolic functions were being played by decision making active inference in the discrete setting while more motor behaviors could be played out by motor active inference in the continuous time setting and through this conversation i'm seeing whether that is explicit or implicit that decision ai to motor ai handoff is mediated by a symbolic layer either implicitly through the code construction or potentially even explicitly with the kinds of approaches that we're exploring and then one last area of some parallels that i'll like to hear from adam or anyone else self-organizing maps brought up by bruno have a lot of similarities with some of the graph operations that tim ferbell and ozankittel and adam's work was describing and then the harmonic modes that adam has described in work as well reminded me of the ultra stability that matt brown described in some of these more classical models of like synergetics and multi-scale harmonic adam or anyone else would be happy to hear any of those thoughts um okay just uh quickly to to answer adam um i don't know exactly i mean what do you mean with oh by the way they can't see my question on the audience maybe we should read ah sorry uh adam is asking to what extent can self-organizing maps be used as model of experience dependent plasticity as implementing a kind of implicit neural architecture search i mean i don't know exactly what you mean with neural architecture search but um um there is this this um literature on growing self-organizing maps so like the usual uh original architecture is is fixed but there is there is some implementations where you actually can add nodes and and they become more more well they have more plasticity and there is also the ones that don't stop learning so it's it's continuously moving so that helps you to not have a fixed structure and a fixed mapping of the sensory input but it's it just moves as there is new data coming new input coming in so that's that's two dynamical systems and growing sums i think if if you are interested in that okay extremely uh i guess the the intuition or question was that um it's like you know i'm sure there are plenty of inductive biases that we come equipped with to help us converge on efficient regimes of inference and learning but i'm also wondering the extent to um i don't want to go too far with this kind of analogy but something kind of like a field programmable gatorade but like that self-organizing map like dynamics could allow for the creation of different regimes of active inference like like to what extent as you're like let's

say um building up a hierarchy at different levels of it based on where you are in this overall hierarchy or hierarchy you might take on um very different computational properties um via experience kind of plasticity so it's like you you learn the inductive biases you need as go um to come back around with self-organizing maps um the locality and the the um the topography of them one thing i'm really curious about is the extent which they could actually be used for um modeling like um enter rhino and hippocampal representations to what degree the different operations that are um used for constructing these maps would also converge with them the kinds of structural inference and structure learning that have been described for the hippocampal system so i don't know the answers to that but it seems like a potentially fruitful intersection that daniel suggested so yeah definitely yeah i mean i think in principle they they are inspired from the hippocampal activity you know the the maps yeah there's lots of proposals out there of different ways of moving them but uh yeah i think they they would be very useful for what you want that's why i'm talking about that like a general question for me that i'm i'm i've been wondering about so i guess i'm suggesting that like you're getting these um represent these representations structure representations potentially being um afforded by these uh place fields of the chemical system one thing i'm wondering is so and that these would provide like equilibrium points like carl often we'll talk about the importance of structuring action and um by these discrete acts that seem to be unfolding at roughly theta frequencies and this seems to be like the time scale of coherent action selection for an organism of about our size that can do about the things we can do with a brain about this big all kind of brought in the same temporal register and so this idea that like we might be stepping along these representations in some ways like moving from like between these bump attractors as somehow structuring cognition by some sort of representation and i'm excited by that but i still don't have like a sense like uh for how far you go with that or ought to go like for instance so let's say we're talking about something like uh um uh this is actually something uh uh carlos that scolded me about in the past which is like before you like uh think of the architectural principle think of the inactive uh situatedness of the system and so something like i wonder is like so you know i'm focusing like maybe you're getting this sort of inductive bias of like the hippocampus rhino system says high level controller but to what extent are we implicitly doing things like symbolic reasoning but via like modes of enactment like to what degree like could you do something like like let's say we're kind of like purling causation and you want to do something like a do operator you could have some sort of like graph some sort of like flow of inference like um maybe even like within like the like chain bump attractors

like the hippocampal system and uh actually another thing that carl actually brought up recently was um bump attractors and the colliculus so maybe even like in terms like patterns of like ocular motor motion is potentially like providing like another means of accessing such representations and being governed by them uh to to what extent it is um is it too neurocentric is it actually like being realized more implicitly by the overall contextualized functioning and like it's you know having open questions for me so sure to carl and then jakub or anyone else who wants to talk to me can i tell adam off again or not i don't think he needs turning off um yeah it's interesting isn't it um i wonder whether that is the um the interface that daniel was referring to um how do you get from a necessary continuous um bayesian filter coupling with the world i mean even with um carla and andy you know these things are moving continuous states better so at some point even if it's actually um physicalized you know there is a continuous uh generative model or um self-organization in play and i'm reading um adam's question as you know is this too neurocentric to to worry about bump attractors and the role of lateral inhibition for example in carving up receptive fields that could be actually written down as just sort of basically a quantized representation you know for example um receptive fields in v5 or motion sensitive areas can either be expressed as a sort of continuous preference for motion as a continuous uh you know function of visual velocity or you can say no this population is just encoding the probability that this particular edge is moving at eight degrees per second so you know if adam you were asking you know um are we being too neurocentric in over interpreting the neuronal dynamics that just basically do the carving up into a series of either tilings and receptive fields or place fields or in the context of some stable fixed points you know sort of um thinking about heteroclinic channels as you know one way of articulating a discrete set of ordinarily organized uh fixed points that have you know that's the way that we should understand it i i don't know but i think you know it could certainly if you can read them in both ways i think that then gives you license if you want to simulate and reproduce and silicone these kinds of computations it certainly allows you to replace heteroclinic channels um heterogeneous cycles uh bump attractors receptive fields of this kind with a discrete representation and you know practically that would be an important and important way to proceed that's simply because that minimizes the complexity of the genetic model and it minimizes the complexity you maximize the uh the model evidence um so you know i don't know whether being neurocentric as a good thing or a bad thing it you know interesting just think about so what is what what is a self-organizing map um so when i was at school it was just basically um a uh a couple map lattice with with lateral inhibition to to um to do this sort

of uh carving up and segregate those various populations um or localities from other localities so if that's the case again you've got this notion of natural inhibition when i take all light dynamics what is that well is that a statement of this exclusion principle and i mean beyond simply your the the sort of exclusion principles i can't be in one place at any one time but the fundamental exclusion that you get when dealing with discrete state spaces you know i've got to be in this state and i'm not in any other state and that means that being in this state means i have to inhibit being in any other state just to get a sum to one constraint so just by going um if you like discrete you're naturally inducing by physical architecture that requires the kind of lateral inhibition that is one of the defining features of a self-organizing map so that might be a virtue uh or you know of of being a bit neurocentric in the sense of you know understanding evolution as working its way to a coarse brain minimally complex discretized symbolic representation of the world and we just now read receptive fields in a slightly oh we over interpret them they're just you know basically encodings i the world is likely to be in this state from my point of view can i just make one final point um which kept on recurring to me that there were people chasing you on this i'm thinking about bruno and self-organizing maps and adam in the uh and actually matt as well with his um i think there's a slow realization that these kinds of architectures are the way to go that's coming out of machine learning when people are getting bored with that propagation so they're now all turning to local energy based rules and they now realize that they get much better performance from a deep neural network if they just use a local energy based rule and what would that look like it would look exactly like the self-organizer self-organizing maps all three of you been talking about it's just me my note trying to predict everybody else around me um uh you know and doing so by minimizing my local variation free energy or minimizing the prediction errors uh you know mathematically those being equivalent you know the twist here is that the you know this can be applied to effectively uh you know overly expressive amorphous a completely connected neural network and it will still work with the right kind of pruning you will start to get to the sort of nice hierarchical structures that bruno was showing uh or the you know with the right kind of temporal scheduling you'll get to the nice that you're the nice uh tolmaneken bound machines but at the end of the day these are just um they're just sort of deep networks or even deep that is networks with a local um a local energy and those things i've now noticed certainly in the university of oxford are now called so there's no necessary no prediction there's no there's no prediction in the temporal sense that adam was trying to emphasize or in the sense of a kalman filter it's just that they rely upon the local computation of prediction

errors to generate the local energy function so i see you know the people the reactionaries in deep learning you know the young turks who want to do it better they've all now identified predictive coding as the rhetoric in order to say this is how it's done and it's just a statement of or your you don't need to do um i think bruno had a nice phrase do you or perhaps it was perhaps it was um france fossil diffuse your learning everywhere through back propagation you if it's working properly you should just be able to do it locally with your own little node vertex of your som or gm in a in an in your fonts if i could uh build on that and then anyone else who'd like to add a challenge that exists and it's recognized in robotics and many of the presentations brought up was how to move from continuous and high dimensional data like a high resolution video camera or multi-sensor integration eventually towards logical and conditional contextual and so when we see the neural network representations we kind of see sometimes zooming in and zooming out and different ways that people are training those models including limitations and those models training like the back propagation that carl just mentioned as well as that stephen grossberg has gone into trying to contextualize that issue in light of what we're discussing about discrete time and about symbolic logic it seems that through the partially observable framework not even saying the partially observable markovian framework but here including holographic principle and so on there's a way to move from a continuous statistical distribution and make a map to a discrete statistical distribution like the a matrix in the pomdps that we use can have a continuous variable and then map that onto a discrete statistical distribution i wondered if there's something like a logical a another tale of two densities a tale of two different cities or a third city and so this matrix recognizes logic from discrete distributions that's the recognition density and then the generative density is the emission of logics that are compatible with discrete distributions because with the recognition direction recognizing conditional logics and searching over relatively small sets of possible conditional logics from discrete distributions is possible like a verb is never used two verbs in a row are not used or something like that from a discrete distribution logic can be extracted or recognized if there's a discrete distribution or more logics can be sampled and that speaks to that inductive bias where you want the inductive bias to have limited false positives and false negatives with respect to the empirical world like you wouldn't want to waste time spitting out logics that cannot be however that might not be lethal but it certainly wastes a lot of time and then conversely what could be lethal would be to have an inductive bias that fails to recognize empirical aspects of the world and so that is like an expectation maximization but transposing not between a continuous and a discrete statistical

space but rather between a statistical and a logical space so jacob or oh yes or bruno anyone who would like to add to that please no jaekov you haven't said much please but yeah i can say something afterwards um yeah i just had some general uh questions that are kind of related to daniel's remark on the move between the purely continuous and statistical representation of the generative models into the discrete and since this week i've been reading the paper on from chris fields and carl um and others on the fep and neuromorphic development i'm wondering whether the question of structure learning can be solved by some fundamental computations that are available at the very low um low dimensional level of just quantum quantum systems where given the right environment that system would naturally evolve into some kind of discrete morphology that it didn't start with kind of similar to like the primordial soup then evolving into discrete uh or more discrete life forms and whether and whether that might impose the the nes the necessary um structural learning principles where message passing where each le each new layer that evolves in the hierarchy is receiving observations from the other ones given by the new markov blanket that's drawn around it but then in terms of actual implementation i uh i'm wondering whether that requires quantum computers or whether that can be done with some kind of different architecture that is able of this uh automatic um uh evolution in its morphology also bruno and then anyone else you'd like to add um yeah very i mean very nice your your your question and then um just kind of to round up what what you were all concerned about um we were just thinking and remembering this this and going back to sums and there is this uh proposal some songs but i don't remember the authors now but it's this other type of architecture that takes into consideration the the learning progress so previous activations and and some sort of history of the winning previous winning uh notes and so on that might be the next step so that you can actually monitor what's happening over time and if you have one of these that it's coding for error then that would include the the monitoring of of the prediction yeah and and it would kind of close this gap between continuous and discrete representations because then you could you could have it all over time so i like if you one of them i can send you some some literature on that that thank you bruno one point on the morphological computational angle uh jacob and then again anyone can raise their hand i thought about a bunch of tree seeds we're back in frequentism again so everyone can take a breath but you know there's a hundred seeds that are planted and depending on the priors inherited from evolution and the updates which are happening from the generative process from the niche there's gonna be like a distribution of that tree through time so like a palm tree is gonna have a very like narrow cone some sort of distributional cone across those

hundred that are sampling from it and then another a shrub might have like a different um shape then you brought up how potentially quantum or discrete like morphological decisions could realize that continuous probability distribution at the population or even like the multiple worlds level and there's something there also with the local computation of large models that carl mentioned that it's like something happens where there is not even just a symbolic or a discrete decision that's made there's an embedded decision that's being made that now is part of the history and so the branching pattern of any given palm tree is going to be unique or any shrub yet they also may fill out at the population level a distribution set that has these aspects that can be modeled as like a gaussian blur over tree morphologies yet of course no one is saying that the tree is a blur and so there's so many interesting contrasts with like the realized trajectory of the lego robot and that's the n equals one the population of trajectories the imagined set of trajectories so having a unified ontology to be able to talk and have formal concise connections amongst these different kinds of remembered now casted anticipated or imagined futures helps find the patterns across systems that like adam said are essentially the basis of sense making and insight i gotta ramble i guess in terms of this like um motion from a continuous to a discrete regime or even like um drawing analogies from like a quantum to like a more classical there's a in general like i've been kind of like with respect to computational models of consciousness wondering um um a seemingly classical world of experience can emerge from a problem like why are things so precise and or um carl's written some really uh interesting papers with andy clark on um on the bayesian blur problem and one of the suggestions was that like the discretization for the sake of action like in order to act you have to act in a particular way and it's like just the requirements of doing a particular thing at a particular time induces this and so one of the things um and so this i don't think it's come uh competing but in some ways competes for like a suggestion i had all those things compatible because like i was wondering whether basically like um something about the the time scales in which you could get like these coherent eigen modes or like the population activity could achieve these like attracting states would create like the time skills of the formation of these large-scale attractors could create like an in and out of like who gets to inform or not and this could help to like sharpen things up and and like creating these population level attractors but there's another way of describing it in terms of like what lets you act um so in terms of like like these different it's one more comment along those lines i think it's interesting that like so um i've talked to some rodent researchers when you remove the hippocampus you'll still actually get like roughly theta scale large mesoscale organization of brain dynamics

for the rest of the rodents nervous system so it seems like there's almost a um it might be and i don't know if you see it's like degenerate but it's like natural selection like any mode any any way it could catalyze things to help with the coordination to help with the alignment the spatial temporal alignment of um that allow that collab for different forms of synchronization was utilized so like there's like you could get for instance a good amount of discretization just from like potentially just from like learning and interacting with the world that requires you to do this just physically to interface with it you get some of it from like the overall um connectomic properties just like that's the times like um that's what's required for the the you know the overall brain to form attractors that they'll form at a certain time scale and different nervous systems of different size and complexity with different bottle like information bottlenecks might tend to form these attracting states roughly on that same scale and maybe things are tuned for that or maybe they get tuned by experience either as an innate inductive bias or as a empirical uh meta prior learning or learned you'd also get it with things like the local logic of like um how you transition between these different equilibrium points for like highly central structures and so um something is very unclear to me is like the extent i imagine the answer is like all of you above in like most combinations but it's like to what degree were these um inductive biases are they taking the form of evolutionary priors and to what degree are they developmental priors this is still very unclear to me um but one thing that um and there seems to be a tension in uh machine learning in terms of these like you want these rich inductive priors to do efficient inference and learning because otherwise it's hopeless but you pay a price in terms of generalization sometimes and so i have no idea how nature balances to what extent in what cases um all right i have some idea but not nearly enough to feel comfortable so anyways that's my ramble i think we can actually hinge on this and bring it to an area that some of the registrants brought up and also something probably many of us have been thinking about which is multi-agent modeling and i wanted to connect that to what adam was just saying um about what is granted via embeddedness what arises or becomes more possible simply as a function of realized corporal embodiment as opposed to in silico simulation and in our multi-agent discussions we've been differentiating spatial multi-agent scenarios from essentially non-spatial like digital or cognitive so in the spatial multi-agent case the real world the niche the generative process does the work of inducing or preventing collisions like it just cannot be the case that two entities are in the same location whereas in the cognitive case whether it's bird song or a negotiation or verbal communication or visiting websites we can both be at the same website and so then there's sort of a mass parallel stigma

g occurring or a mass parallel real-time architecture where there is some kind of coordination but the coordination isn't exclusionary again because two entities could their thought trains could essentially cross in and out and those trains are like ghost trains that don't exclude each other whereas that couldn't happen in the real world i wanted to ask anyone who had thoughts how does multi-agent modeling come into play for robotics what current issues are facing multi-agent robotics and how can what we're discussing here with active inference play a role specifically in understanding complex i i have a question in that regard uh for matt matt your solution has been applied to a single agent uh do you see the possibility of a homeostat that is composed of multiple agents that will direct their collaboration towards achieving a common goal does it scale to multi-agent situations thanks i hope matt can provide a thought as is working carl and then anyone else where is matt listening oh is he can he speak only at a future point in an expected trajectory i will answer for him then i would imagine uh what he would say is if you just take the the generalized synchrony uh perspective on with the emerging properties of a homeostat coupled to its environment and you couple two homeostats together they will find a mutual homeostasis that just will be a joint synchronization manifold so what you'd expect to see is a coming together exactly um um in the spirit of uh dynamical generalized synchronization there will be as i think daniel mentioned earlier on though the homeostats will be singing from the same hymn sheet under the constraints of what they can communicate um so um so from i think from matt's point of view he's more interested though in establishing a generalized synchrony between a a use case an industrial use and his homeostat as opposed to just letting two homeostats you know shape and design their own little economy indulgence sort of cultural niche construction while i'm talking though that it was a nice uh something very interesting about daniel's question in relation to to your work um i i sort of overheard the little robots telling each other what they were thinking um and i thought that was really important um and simply because um you know speaking to this general question about you know what does active inference or what what are the special considerations that you might want to bring to the table when thinking about a multi-agent um setting i think there are two ways we could talk about ages about this first of all it's having the law of requisite variety across agents and then the story would unfold in terms of natural selections bayesian model selection and that would i think answering large parts of adam's questions about where the inductive biases come from if you cast that the biases as selection biases we're just talking about basic model selection selection amongst what the requisite varieties afforded by ashby's law uh but to make that bayesian model selection work you have

to have natural selection um so you know there's a great story about the importance of um multiple agents um from that point of view from the point of view of structure learning for free as an emergent property of of natural basic model selection or natural structural learning but the other the other thing is um just from the point of view of active inference if you've got multiple agents um what's to stop you thinking about these multiple agents as one big agent where you've cut some message passing between them um and like one big agent with lots and lots of eyes and sensors and the like so i'm asking now when i looked at carla and andy what's to stop me thinking about carla and andy as one agent that has a really um flexible and deployable set of sensors so they've got eyes that can point in different directions so if we translate this into sort of um uh you know controlling drones for example and we have a swarm of drones we can think about that as one agent one robot with lots of deployable eyes which gives you you know an enormous flexibility um over uh you know and possibly complexity of the way that we control our two eyes which are both in front of our heads um so how would you then write down a concerted model for a swarm of eyes where there's one cpu there's one um there's one sort of uh the lego brick controlling all the the drones uh where you just write a generative model with multiple sensory modalities where you needed to deploy action in the right kind of way to get the right epistemic foraging or whatever um but that would entail now um message passing belief updating um lateral interactions and an some between the brains of agents so say oh you can't do that you've actually got to have physically separable robots to have a multi truly multi-agent set up so how can you now work around the fact you you don't have direct message passing and belief updating between the two the brains the the brain the separated brains when you just have communication so provided you broadcast what you believe and provided those beliefs are independent so they're conserved from you know so that they you know what i say in terms of my frame of reference um is meaningful from your point of frame of reference which incidentally for daniel and uh jacob becomes relevant from point of quantum frames of reference here but if we can assume we're all got the same we've all got an alice-centric frame of reference then all of it is required to put together the many brains into one brain um is to have beliefs broadcast just those beliefs that that will that will um that are conserved or shared in the uh the ensembles generative model so you know i don't know where your was carla able to hear andy and vice versa so when did did they actually share a narrative in a belief updating no they did not they they uh the the the speaking was uh the talking was mostly for my benefit so i could see what was going in their mind at the time so when when uh uh carl was saying oh andy is freaking out i knew that the

journal of model had identified the behavior and and and inferred the mindset of the other robot but i i am uh playing with the idea that they can communicate um by just distributed message passing from one robot to the other and what they would communicate is is uh i would consider one agent as as a source of sensation for another agent and and in the same way that internally journal models uh the beliefs of a general model becomes a sensation of of another higher level genetic model that could scale across multiple agents and and that would happen by communicating predictions across agents in this prediction errors across agent as they propagate between general evolves within a single agent and just scale the architecture so instead of having a society of mind within an agent which would also have a society of minds across agent in the same mechanisms would be at play but you're absolutely right we need to have this communication but it would be the same kind of communication within uh across society of minds as they would be within society of mind i'm hoping that they would be self-similar and the same architecture would apply at any scale which actually speaks to jacob's question uh about your mark off blankets and mark off blankets and sort of you know that sparse do you get from from that scale-free separation and you know refined minimally complex message passing i think there's something quite fundamental there about you know the nature of communities and the the scale of the world in which we live and all the organs that comprise me living in that world that that scale free aspect that is defined by the sparsity and the absence of coupling and and getting the messages that are communicated right i i think something quite fundamental about that i got a question adam please so for the internal organism internal case i'm wondering so to what degree do you get um common agency for um societies of minds due to common embodiment but when we're going for a kind of common uh or joint agency maybe even different forms of joint identification or de-individuation or re-individuation of individuals into a collective um to what degree do we need something like um as constraining as a joint embodiment like for instance like i'm thinking of uh soldiers marking and like marching in step or like the ways in which like we kind of like seem to like potentially even like expand and hack our body maps by like sharing like different synchronous modes you know that literature is somewhat contested um i like the literature but um the but to what degree is it you don't need to have that kind of synchrony though it's just like a strong enough shared task is enough to like radically make someone uh make the central thing their role of where they're singing in the hem sheet that becomes the primary attractor governing them and you don't need to necessarily have this this kind of physicality i don't know if that makes sense but like i'm think of the different means in which

you yes do you need something like synchronous in time or is it just a very strong pressure selective pressure for coordination to establish a synchronization manifold to create to pull off the re-individuation oh it's a great question adam so thank i also think this ties to again js framing of society of mind and then we can contrast that with a society of bodies so the is that virtualized case whether it's the counterfactual virtualizations that cognitive entities can be modeled as doing or whether it might even be a society of minds in a digital setting and then there's the societies of bodies which maps earlier to our multi-agent discussion on like a physical crowd so like what are the similarities and differences with the digital swarm and a physical one it comes down a lot to their ability to collide and again the digital swarm can weave in and out because they can coexist in the same location in a semantic space in that generalized foraging way whereas the embodied swarm is going to have collision prevention by virtue of the physicality and i think by using active inference to study these systems and integrate them we do approach exactly what you brought up adam which is like almost what do we get through shared tasks as opposed to what do we get from a common task something that we merely have in common versus something that we're actively coordinating on together and potentially even coordinating together in the same exact space or on the same instance of which provides the most constraints like if people are rowing the same boat versus rowing parallel boats versus on different lakes so as we sort of separate spatially and especially virtualize there becomes more and more possibilities um and this for those who are coming for perhaps from a robotics angle and wanting to understand what active inference brings to the picture here hopefully we've pointed a few things like the scale-free or scale-friendly nature of models to be composed based upon their sparsity and defined interfaces and also the multi-modal aspect for example carl asked jf if the audio could be heard rather than only emitted and that's not going to require some bolted on communication module it is going to be able in the symbolic or in the numerical case to be mapped onto an architecture that bruno showed these architectures that can do sensor integration as a function of the interface definitions locally rather than potentially have ad hoc structural design decisions that could result in a lot of research and engineering debt so happy to hear any thoughts on that yes god yeah i mean yeah you're right and um no i just wanted to add that we are working on on that community well i come from from the very i don't know like a very now you could call it old-fashioned school where we were we as you can see in our implementations they are kind of very low level what we could say low level and and coming from the sensory interaction with the world and that that is maybe why we are not or we haven't been so far very concerned with very high level

representations known so we are just dealing with very basic interaction with the world and at the same time it's my belief i mean it's very interesting this interaction between agents and so on but i think or we i've always thought that first we need an agent that that knows itself in the and and has an uh its own model so like a first frontier or a first blanket uh as you like to call them and only then can start to understand other agents and to to interact with other agents yeah so that that was what i wanted to tell you know thyself carl there is an argument from um developmental neural robotics that to have a true sense of self you you've got to have a true sense of other and you can only ever have a true sense of other if there's something else out there that's physically very much like you so just i'm speaking now to adam's uh point does it have to be uh or perhaps daniel's question you know the the shared body as opposed to the shared the shared narrative you there there is an argument uh i'm not making the argument i'm just saying that there could be an argument that in order to disambiguate the causes of some sensory consequences of an where the action could be made by you or an identical robot or me or mum or me or my brother that is the only case in which you are now going to be a need to contextualize and assign and attribute the agency of this outcome to what to self versus other in a world in which i was the only object or phenotype like me i would need a sense of self and i wouldn't need a sense of other it's just when you have multiple agents that are quite similar and confusable um then you actually need a sense of self to make sure it's not it's you and not me um which of course you know speaks to not only theory of mind but you know the essence of communication as well so i'm just making that point that you know um they're putting two robots together maybe as you point out once you've sorted out a robot just learning how to move um then learning that the consequences of it moving could actually be reproduced by somebody else and then it might develop a sense of selfhood a minimal sense of selfhood and then i think you're in the game of true theory of mind that's uh yeah it's a very interesting area with thinking through other minds as have been brought up in some of the presentations and then society of mind it is that nesting of blankets of different kinds of generative models but again using the same statistical or formal machinery just like we could have a nested multi-level regression model where one model was in kilometers and then there was another model that was in degrees for temperature and also the question about swarms and collectives and groups comes up in biology all the time with different ways that people delimit or qualify individuality ranging from the evolutionary unit of replication the kantian concept of an organism as the unit that is teleologically closed the physiological individuality which might even include cases like symbioses the information theory of individuality and

different ways that we can look at information processing and transfer in terms of what individuals are it's so fascinating how that comes into play with robotics where when we think about the robots that will be interacting with us as a slightly different set than the ones that might be working underwater on a pipeline there like like matt showed they might be able to interact with a relatively simple generative process that doesn't exhibit mind-like qualities it's interacting with a mere active inference entity but when on the other side of the blanket there's another adaptive active inference entity and especially when it's us then there's a space of norms and also laws which are like that pseudo code execution order for thinking through other minds in a way how will robots navigate a space and how will they be able to make abduction occur in a real-time way both in the generation of novel hypotheses with inductive bias followed by the selection trajectories of action again taking into account formal codified law as well as the kind of state of exception even or understanding when preferences allow for something to be so strongly desired how and when it's uh pursued might not be how you pursue it in a different time so there's just a lot of cool threats yes adam and anyone else um this is reminding me also from the uh first session uh uh there was mention of value alignment and like this seems to be like a multi-scale problem in terms of like it will show up just as much of like the robot not stepping on your toes or you know crushing your foot to like um as it expands you know does it uh is it is it optimizing in the direction are its tensors going the direction that you want your tensors to go of shaping um and it seems like of the proposals sometimes how did i say this like begged questions that active inference may be answers so it's sort of like let's say you have a framework like cooperative inverse reinforcement learning and so there's some sort of like you know so it's like you are trying to like roughly like optimize for the utility function of the other agent so how do you get this kind of like shared utility but in like these like active inference models where like you're bootstrapping minds um like intersubjectively you have like regimes of joint attention and thinking through other minds or i mean some of this work with them uh i'm doing with anna uh chinitza it's like she emphasizes how like we start out like actually physically inside of another organism completely dependent like homeostatically and so like you're you're automatically sort of um grandmothering in this joint intention right from the get-go the seeds of um you know organization and individuation are already like you know scaffolded in that way just by the um just by by the niche contextualization where you are co-constructing each other as your mutual and you and me are inter-subjective like mutual niche so it seems like um with respect to the um i don't know at what point these problems in terms of like deploying robots when they first

enter because you know it's it's all you know always highfalutin as like intelligence explosions really like a recursively amplifying system stay aligned with you but just like you know how can i make sure the robot doesn't crush my foot like i don't anyway so um the ways in which um co-valuing and um the the robot identifying itself as an individual relative to other individuals plays into it it seems like active inference has like a very rich way of handling and modeling a lot of these different um uh proposals and processes but um something i wonder like from to ask like actual roboticists like how near-term is it like a live problem of getting like what forms of social modeling for robots like and what forms of like a sense of self modeling for the robot like how much do we need at what stage for what levels of deployment in the world with what degrees of robustness simple question great question and maybe just uh to double on what active inference does to potentially frame that junction since in this round table we won't address but this is the stuff and substance of applied active inference symposia for many years to come in an economic framework a reward absolutist framework which is kind of a full stack ranging from the ways that models are trained based upon pragmatic value and reward all the way up through a world that values economic returns the question becomes in human robot alignment or in multi-agent alignment more generally the question becomes how are we creating and then allocating epistemic i'm sorry how are we creating and then allocating pragmatic value and epistemic can be valued to the extent that through time it provides pragmatic value for example one person doing a little bit more of a research angle and one doing more of an application and profit side of the business and when we shift to a uncertainty imperative for a sustainable or even optimistic world model then we can achieve pragmatic value as a consequence of a process theory that highlights reduction of uncertainty and then the question moves from creating and allocating value how can this ecosystem of diverse entities find a general synchrony and communicate be how they are in a way that will be reducing the expected free energy of the ensemble which is not the question that was even approached by the create value allocate value framing that has come up in ai and continues to occur to this day what if the image creators what if they end up reducing our food allocation so they can have more they take the intellectual property instead of providing it to humans that might be framed slightly differently in an uncertainty reduction framework plus optimism rather than in a reward maximization which somewhat enforces a atomism and even a pessimism that was an excellent point yeah there are so many ways you can tackle that and i'm still puzzling over adam's charge how do you stop robots standing on your foot i was thinking how do you stop little puppy dogs wing on the carpet um you know i i think it's really going to

train them haven't you really i don't think there's any magic answer i think the answer is just what daniel said you know uh things are designed and um can presumably only exist under the um if they um minimize their uncertainty or expected surprise so that they make everything as predictable as possible which simply means that if i'm living in a world full of dogs robots cats flies and plants um then we're all trying to make ourselves as predictable as possible to each other and that is the you know that's a sort of folk psychological way of saying that variation free energy is an extensive quantity so if a set of um markov blankets or phenotypes of varying forms and structures jointly has its own markov blanket so it is an ensemble or a group or a community or a family um then it must put an upper bound on its surprise and free energy and because the free energy of each constituent of that ensemble summed to give the joint free energy then it also requires that each individual is trying to minimize her surprise at the same time it's all internally consistent and quite you know to my mind unmagical um if you just look at it in terms of minimizing free energy or minimizing optim surprise under optimism was telling all that supported thanks you'd have to make your dog surprised when he wheeze so you've got to find a way of making the robot surprised when it stands it stands on your on your foot jf you have a dog right so how do you see similarities and differences with these versions of and also potentially even a family so how do you see these different entities being um communicating and and behaving appropriately or not and learning well that's having a dog it raises the question of do i make myself understood and that's a very that's a very difficult question to answer when i say no you're not allowed to eat from that plate on the table i think my dog understands i'm not allowed to do this now as soon as i walk out then permission is granted so i you know it's it's difficult to communication is very difficult across across individuals especially across the species but i'd like to to just just rant a little bit on something came up which is uh the scale-free nature of free energy principle the fact that it can apply at a societal level the same way that it applies like a russian doll to the cellular level as a software developer i see self-similarity the recursion fractal as as a as a beautiful thing and i think at the same way physicists have symmetries as as a guide to what is right because it's beautiful i think we share um self-similarity as as a guide as what is beautiful and possibly true and i think self-similarity is beautiful because it's it's it's it's collapses the amount of information that you need to generate something rich um so that's why i think that having a society of mind made up of actors and having a society of minds uh and having the same principles scale up uh feels very beautiful and feels right excellent um yes just if we could go for any any length of time a final set of thoughts and reflections from each of you quick

interjection uh jf if you have anything written on that or want to write it there's an upcoming special issue on symmetries that i think it might fit with would you like to give any final reflections or anything that you've heard or felt updated by any thoughts you have on the state of the applied active in 2022 in robotics or where we're heading now that this symposium is behind us i'll start with adam but then everyone else can go thank you daniel well this simulated a lot of questions and just seeing the range of work being done was inspiring and um all i have to say is i was grateful to be a part of thank you for putting this together thank you adam and after the closing words i'll review every presentation and list the co-organizers as well so bruno and then we'll continue on yeah just just like adam said thanks a lot for the invitation it was it was very kind of you i think i want to thank mark for insisting mark miller that was insisting on us to present with you here um it was it was a very nice experience and well we keep we we keep working we need more hands we have more ideas than hands but i suppose that's everyone's problem no so great then there were three uh-huh jf yes well again thank you very much this was extremely stimulating every talk i came up i came out listening to it with this i want to introduce in my robots some aspect of it uh and my mind is is buzzing right now with with possibilities and so thank you so much for this opportunity i guess uh i can go next also thanks a lot for uh um being able to join this this final round table and for all of the great presentations i'll definitely have look over them probably a couple couple more times uh to fully uh understand all the progress that um has been has been made and specific specifically on there's a lot of a lot to consider on the multi-agent uh modeling and the ideas uh in this discussion that we were also talking about quite a lot in the active block friends uh project probably was left with uh more more questions at the at the end of it but that's also the exciting part of it so thanks a lot and now it's my turn i cleverly leave myself to last because everything that needs to be said has already been said so that just means all i have to do is just to thank daniel and his colleagues which he's going to thank anyway but personally to daniel thanks so much for all your energy not just today but over the years it is really great to see these people and coys come together and sort of you know generate ideas and questions which is exactly what we need to do thanks a lot carl and i think by one personal reflection it swooped in at the end when jf said that at from each of the presentations there was like a module or an archetype or a function that transposed and i think that is proof of concept of working in a shared ontology as well as in an embodied shared ontology which is to say like a field in a community of practice with different small world structures and so on but when we have the ability to transpose across systems and scales then we can bring in the modules and

the pieces that people are mentioning and start to understand how those functions can be composed and designed and that's extremely exciting everyone thanks for joining the round table you can depart the zoom or i'm just going to review the uh presenters again and the co-organizers so thank you for joining the round table you can leave if you'd like in the first session there was tim schneider with active inference for robotic manipulation tim verbellan robots modeling the world from pixels using deep active inference ben white artificial empathy active inference and collective intelligence norse jeet learning agent preferences and wen hua chen dual control for exploitation and exploration and its applications in robotic autonomous search then in the first round table it was wenhua and i talking about dual control the second session began with bruno lara with prediction implementation matt brown real-time robotic control through embodied homeostatic feedback and atom saffron with generalized g-slam as unification framework for natural and artificial intelligences towards reverse engineering the hippocampal enterinal system and principles of high-level cognition and then jf cloud here active inference for lego robots we've just completed the second round table with adam bruno jf carl jacob and i the organizers for the symposium were mark miller matt brown blue knight alex vyatkin yvonne metelkin and myself thanks again till number three of applied active inference hope to see everyone around the institute bye